

Additional Practice

A1.5.01

Problems 1–2: Here is a shape puzzle. The sum of each row and column is shown.

1. Select *all* the true statements.

☐ A.  +  = 22

☐ B.  +  = 14

☐ C.  +  = 15

☐ D.  = 12

☐ E.  +  = 19

		= 17
		= 24
		= 19 = 22

2. Show or explain why this statement is *false*:  = 8

Problems 3–4: Here is a shape puzzle.

3. Determine the solution for this puzzle.

Shape	Value
Heart	
Flower	
House	

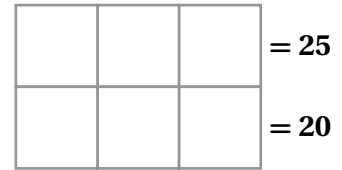
			= -3
			= -3
= -7	= -8	= 9	

4. Explain or show your thinking.

Problems 5–6: Use these two equations:

$$x + y + y = 25$$

$$x + x + y = 20$$



5. Draw a shape puzzle to represent these equations.

6. Determine the values of x and y .

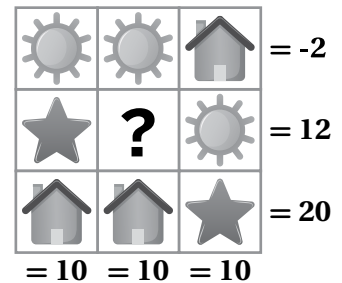
$$x = \quad y =$$

Problems 7–9: Here is a shape puzzle.

7. Determine the missing shape in the center of this puzzle.
Circle your choice.

Sun Star House

8. Show or explain your thinking.



9. If the missing shape has a value of 8, what are the values of the other two shapes?

Additional Practice

A1.5.02

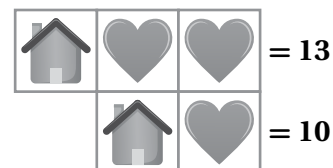
1. Solve this system of equations. Use the shape puzzle if it helps with your thinking.

$$2x + y = 13$$

$$x + y = 10$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$



2. Which equation is the result of adding these two equations?

$$-3x + 2y = 1$$

$$3x + 5y = -15$$

A. $7y = -14$

B. $7y = 16$

C. $-6x + 7y = 16$

D. $6x + 7y = -14$

3. Which system of equations is solved in the fewest steps using elimination by addition?

A. $-x + 4y = 5$
 $4x - y = 12$

B. $-4x + 3y = -16$
 $-2x + 3y = 12$

C. $5x - y = 7$
 $5x - 2y = 14$

D. $-6x + 2y = 12$
 $6x + 7y = 3$

4. Which ordered pair is the solution to this system of equations?

$$-2x + y = 11$$

$$2x - 3y = -25$$

A. $(-3, 8)$

B. $(-2, 7)$

C. $(2, -7)$

D. $(4, 7)$

Problems 5–8: Determine the solution for each system of equations. Show your thinking.

5. $4x - 7y = -3$

$4x + 7y = 67$

6. $5x - y = 4$

$-5x + 3y = -3$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

7. $-x + 8y = 18$

$-3x + 8y = 6$

8. $5x - 3y = -26$

$5x + 4y = -12$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

Additional Practice

A1.5.03

1. Select *all* equations that are equivalent to $2x - 9y = -5$.

☐ A. $-2x + 9y = 5$

☐ B. $4x + 18y = 10$

☐ C. $-6x + 27y = 15$

☐ D. $6x + 27y = -15$

☐ E. $4x - 18y = -10$

2. Diondre and Gia are solving this system of equations. They disagree about what the first step should be to eliminate a variable.

$$5x - 3y = -13$$

$$-10x + y = -21$$

Diondre's strategy: Multiply $5x - 3y = -13$ by -2 and then add the equations.

Gia's strategy: Multiply $-10x + y = -21$ by 3 and then add the equations.

Whose strategy will eliminate a variable once the equations are added?
Circle your choice.

A. Diondre's

B. Gia's

C. Both

D. Neither

Explain your thinking.

3. Select *all* of the following equations that are equivalent to $-2x + 12y = 8$.

☐ A. $4x + 8y = 48$

☐ B. $20x + 120y = 80$

☐ C. $x - 6y = -4$

☐ D. $-10x + 60y = 40$

☐ E. $-2x - 12y = -8$

Problems 4–5: Determine the solution for each system of equations. Show your thinking.

4. $-2x + 9y = 24$
 $8x - 3y = -30$

5. $-x + 4y = -30$
 $3x - 2y = 20$

$x = \dots\dots\dots$ $y = \dots\dots\dots$ $x = \dots\dots\dots$ $y = \dots\dots\dots$

6. Solve the system of equations using the methods below.

$-x + 6y = 21$
 $4x - y = 31$

Eliminate the x -term first.

Eliminate the y -term first.

The solution is (..... ,).

The solution is (..... ,).

Additional Practice

A1.5.04

1. Identify the solution to this system of equations.

$$y = x + 5$$

$$y = -x + 9$$

A. $(-2, 3)$

B. $(0, 9)$

C. $(2, 7)$

D. $(4, 5)$

2. Identify the solution to this system of equations.

$$x - y = -6$$

$$-3x - 2y = 8$$

A. $(-4, 2)$

B. $(-3, 3)$

C. $(-1, 5)$

D. $(4, -2)$

Problems 3–4: Show or explain what your first step would be for solving each system of equations.

3. $3x - 2y = -13$

$$y = 5$$

4. $y = -4x + 12$

$$6x + 5y = 11$$

5. Show or explain how you would solve one of the systems of equations using the substitution method.

System A

$$2x - 3y = 9$$

$$5x - 4y = 11$$

System B

$$y = 4x - 3$$

$$-3x + y = 7$$

Problems 6–7: Solve each system of equations.

6. $3x - 5y = 18$
 $3x = 4y + 15$

7. $4x + 2y = 21$
 $2y = 25 - 5x$

8. Match each system of equations with its solution.

System of equations

Solution

a. $3f - 2y = 11$
 $y = 3f - 13$

..... $(5, 2)$

b. $x + 8y = -1$
 $2x - 6y = -13$

..... $\left(2, -\frac{2}{3}\right)$

c. $4x = -12y$
 $10 + 6y = 3x$

..... $\left(-5, \frac{1}{2}\right)$

9. Bard is solving this system of equations: $-3x + 2y = 16$
 $5x - 3y = -18$

Bard begins by rearranging the first equation to isolate the y variable: $y = 8 + 1.5x$.
 Then Bard substitutes the expression $8 + 1.5x$ for y in the second equation, as shown:

$$5x - 3(8 + 1.5x) = -18$$

$$y = 8 + 1.5x$$

$$5x - 24 - 4.5x = -18$$

$$y = 8 + 1.5(-84)$$

$$0.5x - 24 = -18$$

$$y = -118$$

$$0.5x = -42$$

$$x = -84$$

Does Bard's solution of $(-84, -118)$ make both equations in the system true?
 Explain your thinking.

Additional Practice

A1.5.05

1. The point (2, 1) is on the line $y = 3x - 5$. Explain how you can determine if this point is the solution to this system of equations:

$$y = 3x - 5$$

$$y = x - 1$$

2. Here is a system of equations.

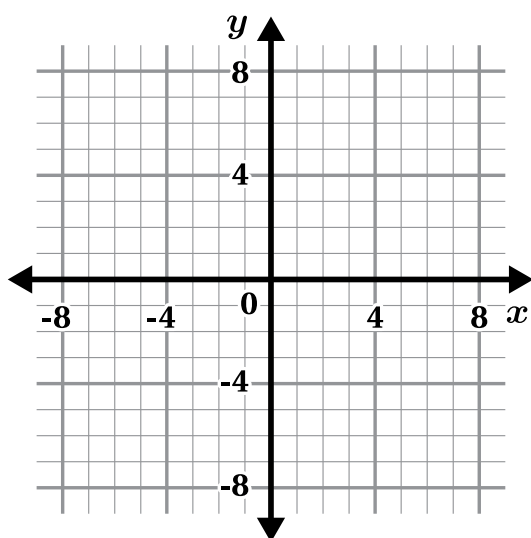
$$y = -\frac{1}{2}x + 1$$

$$y = \frac{1}{2}x - 3$$

How many solutions does this system of equations have? Explain your thinking.

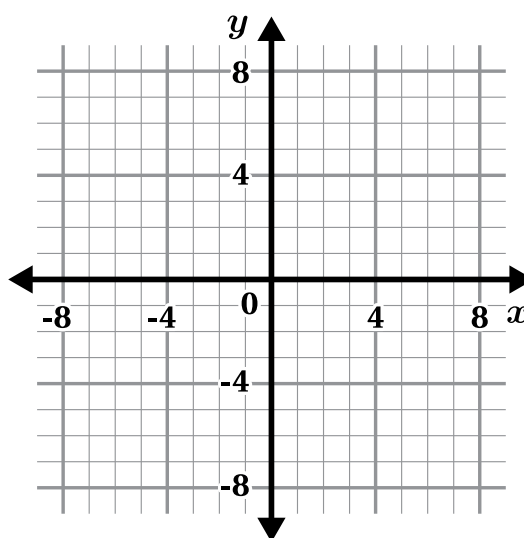
Problems 3–4: Solve this system of equations. Write the solution as a coordinate pair.

3. $y = -3x - 4$
 $y = \frac{1}{3}x + 6$



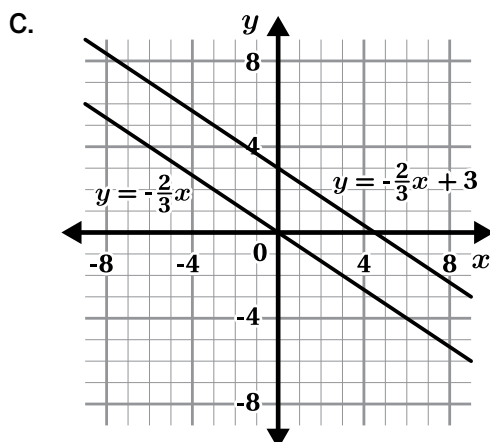
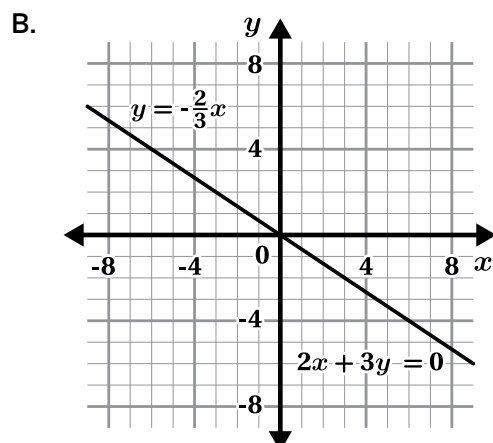
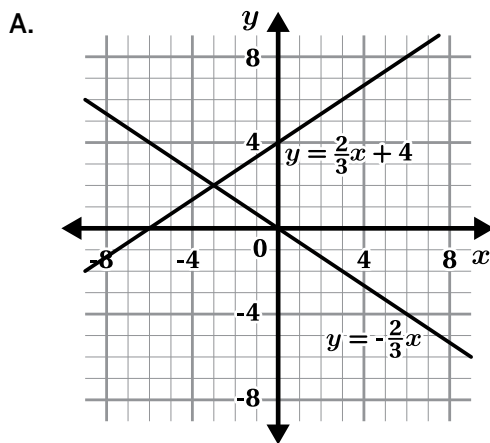
Solution:

4. $y = 2x - 4$
 $y = -x + 5$



Solution:

5. Which graph shows a system of equations that has no solutions?



6. Match each system of equations to the number of solutions it has.

Equation

Solution

a. $y = -3x + 4$
 $y = 3x + 4$

..... no solutions

b. $3x + y = 4$
 $y = -3x - 4$

..... one solution

c. $3x + y = 4$
 $y = -3x + 4$

..... infinitely many solutions

7. A system of equations has infinitely many solutions. Select *all* of the statements that must be true about the equations in this system.

☐ A. The equations have different slopes.

☐ B. The equations have the same slope.

☐ C. The equations have different y -intercepts.

☐ D. The equations have the same y -intercept.

Additional Practice

A1.5.08

Problems 1–3: Show or explain what your *first step* would be to solve each system of equations.

1. $7x - 12y = 20$
 $15x + 12y = -9$

2. $y = \frac{1}{2}x - 8$
 $y = -\frac{2}{3}x + 6$

3. $5x - 9y = 21$
 $x = -6$

Problems 4–5: Determine how many solutions each equation has. If there is one solution, what is the solution?

4. $5x - 10y = 40$
 $3x + 10y = 8$

No Solution

One Solution

Infinitely many solutions

5. $8x - 16y = 18$
 $x = 2y + 9$

No Solution

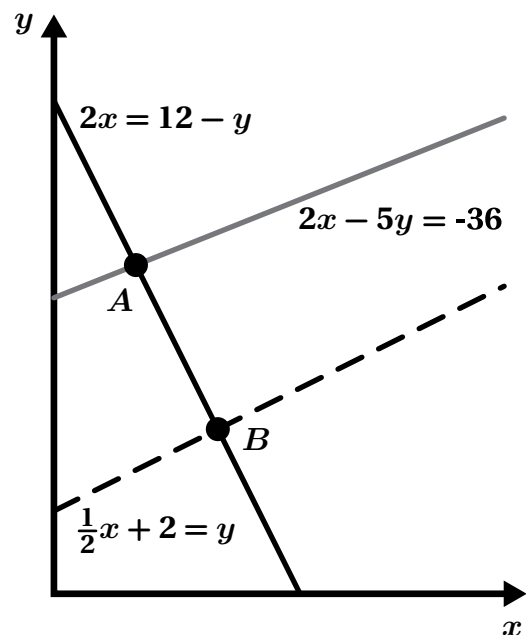
One Solution

Infinitely many solutions

6. Determine the coordinates of points *A* and *B*, the intersections of the lines on the graph.

A =

B =

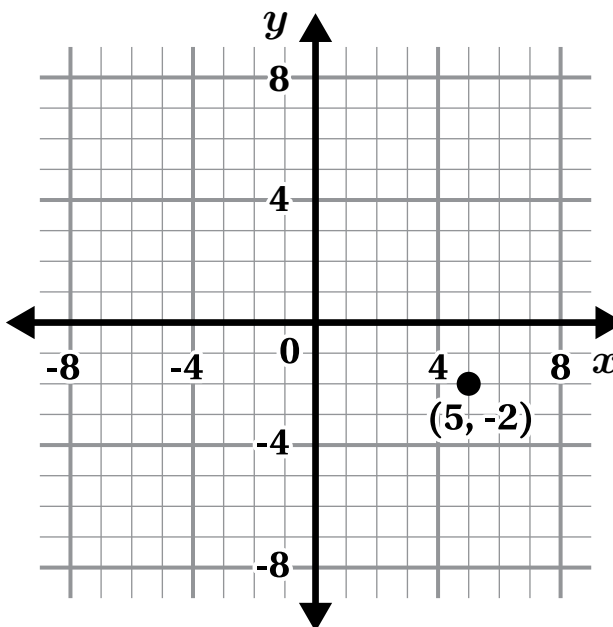


7. Consider the equation $4x - 3y = 12$. Match each description with an equation that would make a system of equations with the given number of solutions. Show or explain your thinking.

Description	Equation
a. One solution	 $4x - 3y = 6$
b. No solution	 $y = \frac{4}{3}x - 4$
c. Infinitely many solutions	 $4x + y = 20$

8. Write a system of equations where $(5, -2)$ is the solution.

Use the graph if it helps your thinking.



Additional Practice

A1.5.09

1. Here is a system of inequalities.

$$y \leq -x + 4$$

$$2x + y > 8$$

Select *all* points that are solutions to the system of inequalities.

☐ A. (5, -1)

☐ B. (6, -3)

☐ C. (0, 0)

☐ D. (7, -5)

☐ E. (-5, 7)

☐ F. (6, 3)

Problems 2–3: The student council has \$300 to spend on burgers for a barbecue. They want to serve at least 240 burgers. A package of veggie burgers, x , costs \$10. A package of turkey burgers, y , costs \$20. The student council wrote this system of inequalities:

$$10x + 12y \geq 240$$

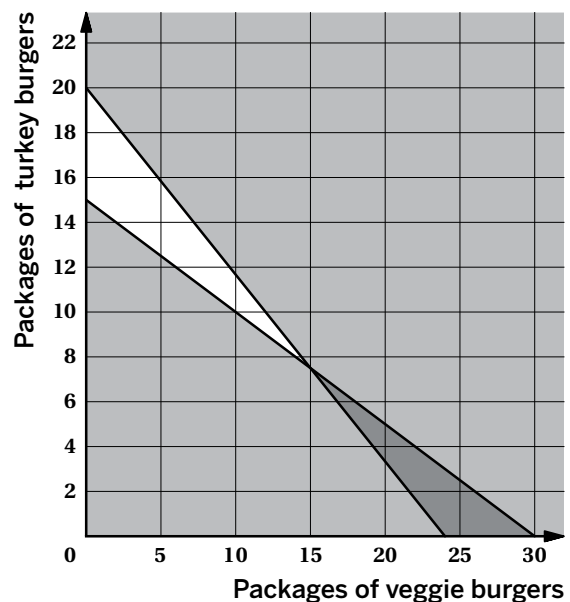
$$10x + 20y \leq 300$$

2. What does the inequality $10x + 20y \leq 300$ represent?

3. Here is the graph of the system of inequalities.

Which of the following combinations of packages of veggie burgers and turkey burgers could the organizers buy?

- A. 0 veggie burger packages, 15 turkey burger packages
- B. 10 veggie burger packages, 12 turkey burger packages
- C. 15 veggie burger packages, 8 turkey burger packages
- D. 25 veggie burger packages, 2 turkey burger packages



Problems 4–5: Andre needs to buy batteries for his business.

AA batteries come in packs of 24 and cost \$15 per pack. D batteries come in packs of 12 and cost \$15 per pack. Andre wants to buy at least 180 batteries and spend no more than \$150.

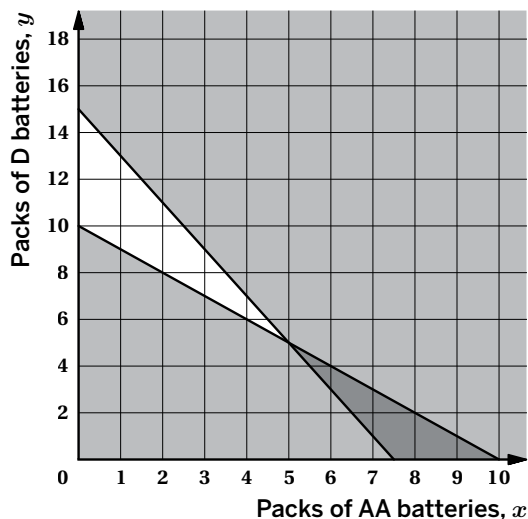
Here is a systems of inequalities to represent Andre's constraints.

$$24x + 12y \geq 180$$

$$15x + 15y \leq 150$$

4. What is a combination of AA batteries and D batteries Andre that meets both constraints?

5. What is a combination of AA batteries and D batteries that meets no constraints?



Problems 6–7: A dog groomer charges \$24 to give a small dog a bath and \$30 to give a large dog a bath. A bath takes 30 minutes for a small dog and 1.25 hours for a large dog. The groomer works up to 6 hours each day, and she needs to earn at least \$180 a day.

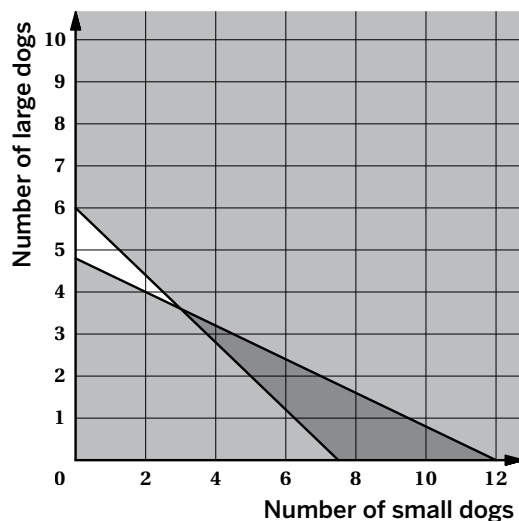
$$24x + 30y \geq 180$$

$$0.5x + 1.25y \leq 6$$

6. Identify which of the following points meets the groomer's requirements. Explain your thinking.

- | | |
|-----------|-----------|
| A. (4, 4) | B. (6, 2) |
| C. (7, 3) | D. (9, 2) |

7. Explain why (8, 1.5) may be a solution to the systems of inequalities but does not represent a combination that meets both constraints.



Additional Practice

A1.5.10

Problems 1–2: Emika graphed the boundary lines for this system of inequalities.

$$2x + 5y \leq 10$$

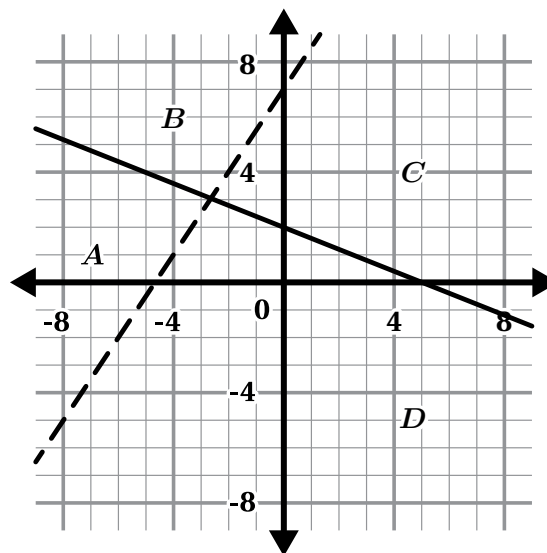
$$2y > 3x + 14$$

1. Which letter represents the solution region to the system of inequalities? Circle one:

A *B* *C* *D*

2. Is the point $(4, -5)$ a solution to this system? Circle one:

Yes No



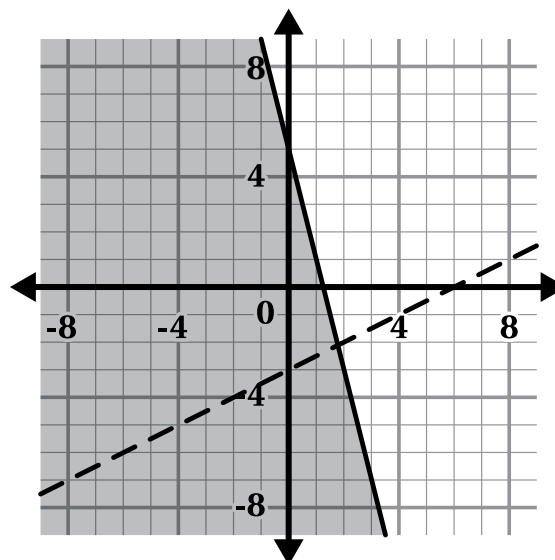
Problems 3–5: Taryn graphed the first inequality and the boundary line of the second inequality.

$$4x + y \leq 5$$

$$-x + 2y > -6$$

3. Complete the graph of the second inequality.

4. Explain how you knew where to shade the second inequality.

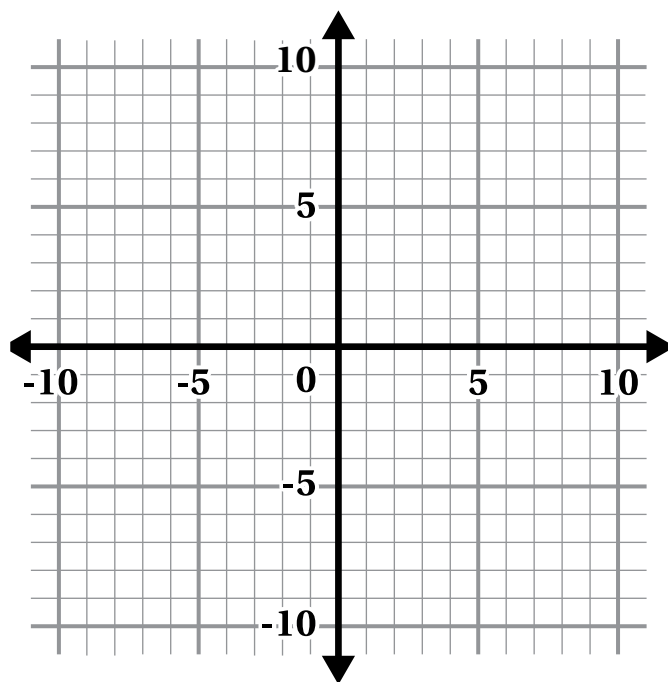


5. Is $(0, -3)$ a solution to this system? Explain your thinking.

Problems 6–7: A coordinate plane is provided.

6. Create a graph of a system of inequalities that has no solutions.

7. Explain how you know it has no solution.

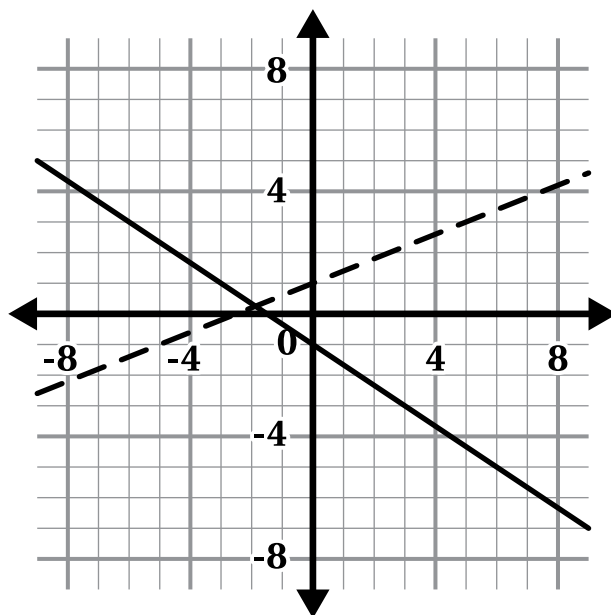


Problems 8–9: Randall graphed the boundary lines of this system of inequalities:

$$-2x + 5y > 5$$

$$2x + 3y \geq -3$$

8. Complete the graph of the system of inequalities.
9. Identify a coordinate pair that is in the solution region.



Additional Practice

A1.6.01

Problems 1–3: Wendy is observing a colony of mice in which its population doubles every week. The table shows the population of mice over time, in weeks.

1. What was the population of the mice at the beginning of the experiment?

Time (weeks)	Population
0	?
1	12
2	24
3	48
4	96
...	...
6	?

2. What will the population of mice be after 6 weeks?

3. Ricardo observed a different colony of mice that he believes populates at a rate slower than Wendy's mice. He wrote this equation: $p = 12 \cdot 1.5^t$. He used p to represent the population of mice in his colony and t for time, in weeks. Explain what the 12 and 1.5 mean in this situation.

4. In the table, x represents the number of minutes and y represents the number of bacteria cells in a sample.

x	0	8	16	24	32	40	48	56
y	4	8	16	32	64	128	256	512

Select *all* the statements that describe the growth.

- ☐ A. Every 8 minutes, the number of bacteria cells doubles.
- ☐ B. Every 4 minutes, the number of bacteria cells increases by 8.
- ☐ C. At the end of the experiment, there are 56 bacteria cells in the sample.
- ☐ D. At the beginning of the experiment, there are 4 bacteria cells in the sample.
- ☐ E. At the beginning of the experiment, there are 8 bacteria cells in the sample.

Problems 5–7: Caroline has a new toy fish whose mass quadruples every hour when you add water. The fish’s weight was initially 2 grams.

5. Select *all* the expressions that represent the weight of Caroline’s toy fish, in grams, after 3 hours?

☐ A. $2 \cdot 4 \cdot 4 \cdot 4$

☐ B. $2 \cdot 2 \cdot 2$

☐ C. $2 \cdot 4^3$

☐ D. $2 + 4 \cdot 4 \cdot 4$

☐ E. $2 \cdot 3^4$

☐ F. $2 + 4^3$

6. Complete the table.

Time (hrs)	0	1	2	3	4
Weight (grams)					

Problems 7–8: Dion is reading a book about a zombie apocalypse. The story began when 10 zombies entered a convention center where people were attending a concert. The total number of infected people increased by 2.5 times each day.

7. Write an equation that represents this situation where n represents the total number of people infected by the zombies and t represents the number of days that have passed since the apocalypse began.
8. Approximately how many people will be infected after 4 days have passed?

Additional Practice

A1.6.02

Problems 1–3: These tables show the number of blue and green globs each day.

1. How many of each type of glob will there be on day 4?

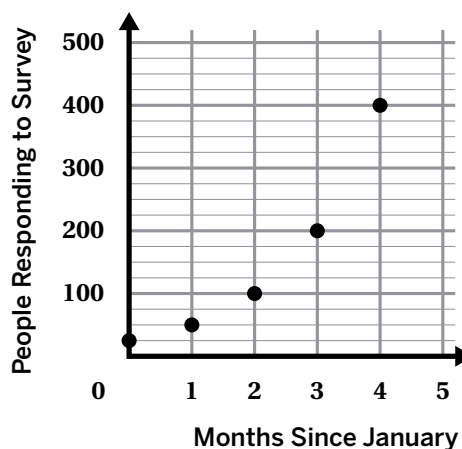
Day	0	1	2	3	4
Blue Globs	4	8	16	32	

Day	0	1	2	3	4
Green Globs	12	24	36	48	

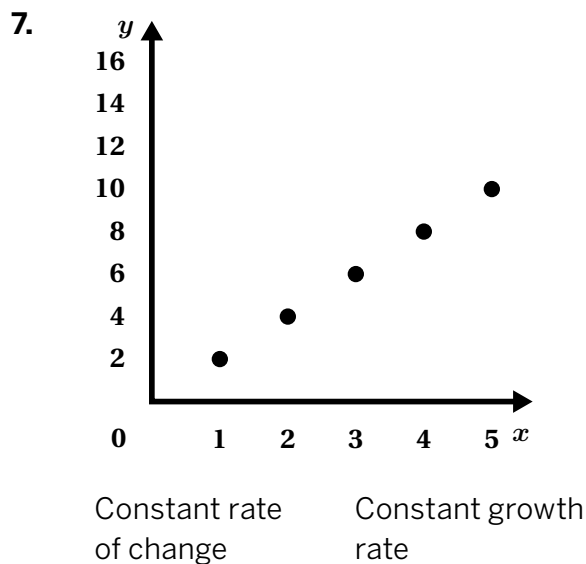
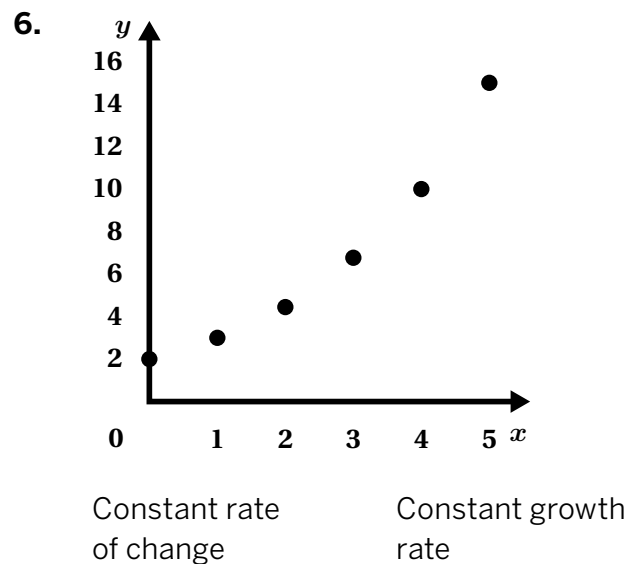
2. Will there be more blue or green globs on day 10? Show or explain your thinking.
3. Which group of globs changes by a constant rate of change? Show or explain how you know.

Problems 4–5: This graph shows the number of people who responded to an online survey about a new neighborhood park since January.

4. How many people completed the survey in January?
5. Does the number of people responding to the survey grow by a constant difference? Show or explain how you know.



Problems 6–7: Determine whether each graph shows a constant rate of change or a constant growth rate. Circle your choice.



Problems 8–9: Determine whether each table shows a linear or exponential function. Circle your choice.

8.

x	1	2	3	4
y	1	8	15	22

Linear Exponential

9.

x	1	2	3	4
y	9	18	36	72

Linear Exponential

Additional Practice

A1.6.03

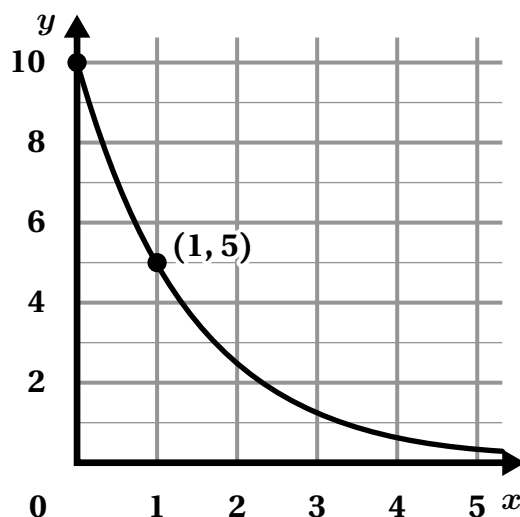
Problems 1–3: Determine the value of each expression when $x = 3$.

1. 4^x

2. $\left(\frac{1}{2}\right)^x$

3. $4(2)^x$

4. Here is a graph of $y = 10\left(\frac{1}{2}\right)^x$. Explain where you can see the 10 and the $\frac{1}{2}$ in the graph.



Problems 5–7: Match each equation to the graph that represents it.

Equation A

$$y = 12\left(\frac{1}{3}\right)^x$$

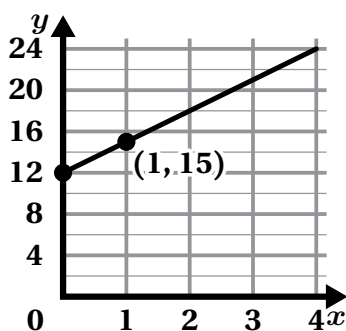
Equation B

$$y = 12(3)^x$$

Equation C

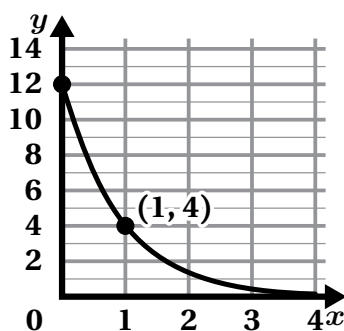
$$y = 12 + 3x$$

5.



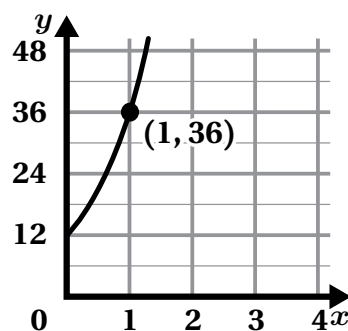
Equation _____

6.



Equation _____

7.

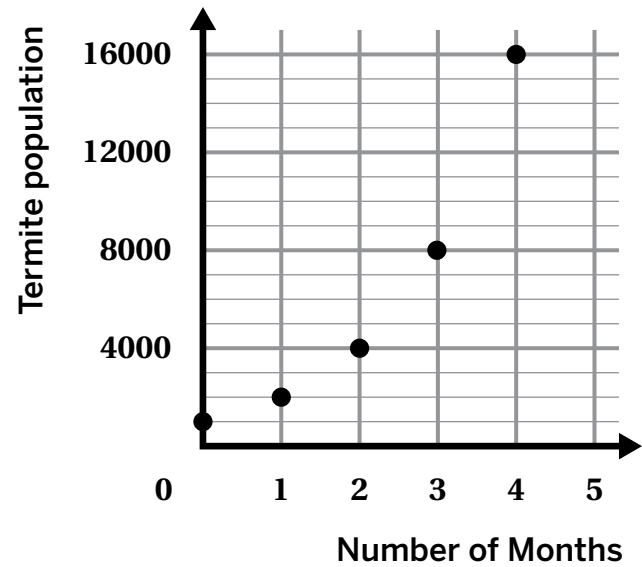


Equation _____

8. Explain how you determined which equation to match with the graph in Problem 7.

Problems 9–10: The number of termites in a colony is measured each month and the results are plotted on the graph.

9. What was the termite population when it was first measured?



10. What equation represents the termite population, t , for m months after it was first measured?

Additional Practice

A1.6.05

1. Which equation best models the data in the table?

A. $f(x) = 48(1.25)^x$

B. $f(x) = 36 + 0.75$

C. $f(x) = 48(0.75)^x$

D. $f(x) = 36(0.75)^x$

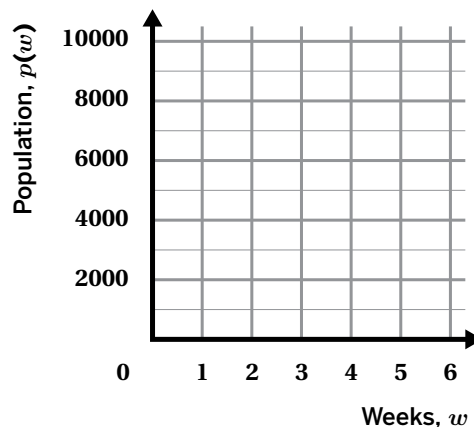
x	$f(x)$
1	36
2	27
3	20.25
4	15.1875

Problems 2–4: The equation $p(w) = 100 \cdot 3^w$ models a butterfly population, $p(w)$, where w is the number of weeks after the population was first measured.

2. Complete the table.

Weeks, w	Population $p(w)$
0	
1	
2	
3	
4	

3. Graph the situation.



4. Where on the graph do you see the 100 from the equation?

5. Determine the value of $p(-3)$ and explain what it means in this situation.

Problems 6–8: A walrus population, $p(t)$, is modeled by the equation $p(t) = 300 \cdot \left(\frac{4}{3}\right)^t$ where t is the time, in years, since the population was first measured in 2018.

- 6.** How many walrus were in the population in 2018? Show or explain your thinking.

- 7.** How many walrus were in the population in 2022? Show or explain your thinking.

- 8.** How many walrus were in the population in 2016? Show or explain your thinking.

Additional Practice**A1.6.07**

Problems 1–3: A group of ornithologists were researching the hummingbird population in a state park over several years. There were 500 hummingbirds when they first started keeping track. The population has increased by 20% each year.

1. How many hummingbirds are in the state park 1 year after the ornithologists first counted?
2. Write an expression that represents the hummingbird population after 3 years.
3. Write an expression that represents the hummingbird population after n years.
4. Last year, the attendance at a water park was 825,000 people. This year, attendance increased by 7%. Select *all* the expressions that represent the attendance at the water park this year.
 - ☐ A. $825000 + 0.07$
 - ☐ B. $825000(1.07)$
 - ☐ C. $825000(1 + 0.07)$
 - ☐ D. $825000(1 - 0.07)$
 - ☐ E. $825000 + 0.07(825000)$

Problems 5–6: Kendra deposited \$1,200 in her savings account with an annual interest rate of 3.5%.

5. Write a function, $f(t)$, to represent the amount Kendra will have in her account, in dollars, after t years.

6. Complete the table to determine how much money Kendra will have in her savings account over time, if no additional deposits or withdrawals are made.

Time (yr)	Amount in Account (\$)
0	1200
1	
2	
3	
4	

Problems 7–9: Three cities have the same initial population and different percent increases each year. Match each function $p(t)$, representing the population after t years, with its correct description.

$$p(t) = 12000 \cdot (1.45)^t$$

$$p(t) = 12000 \cdot (1.045)^t$$

$$p(t) = 12000 \cdot (1.0045)^t$$

7. City A has a 4.5% annual increase in population.

.....

8. City B has a 0.45% annual increase in population.

.....

9. City C has a 45% annual increase in population.

.....

Additional Practice

A1.6.10

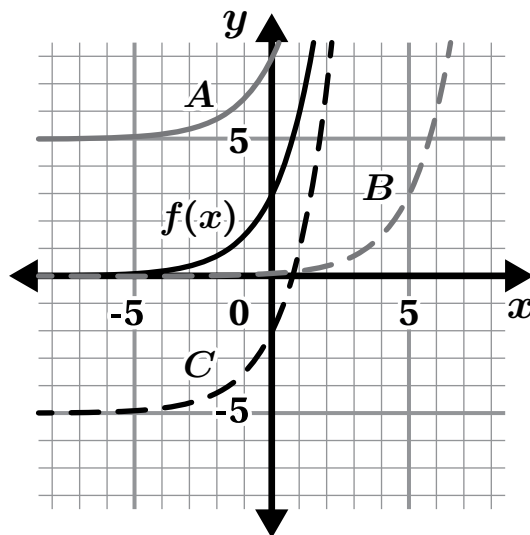
1. Here is a graph of $f(x) = 3 \cdot 2^x$.

Match each function with its graph.

$f(x) = 3 \cdot 2^x - 5$

$f(x) = 3 \cdot 2^x + 5$

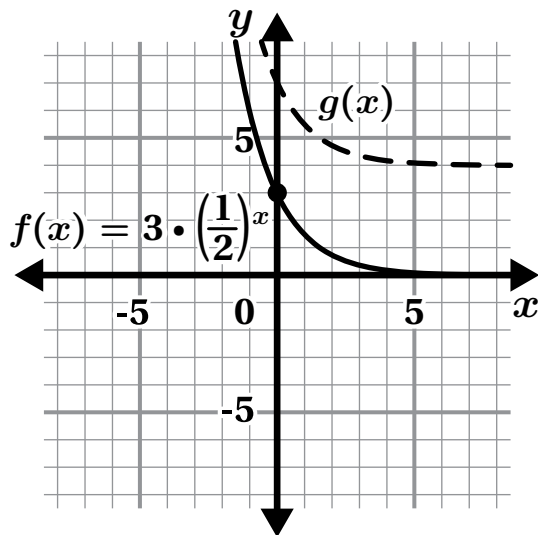
$f(x) = 3 \cdot 2^{x-5}$



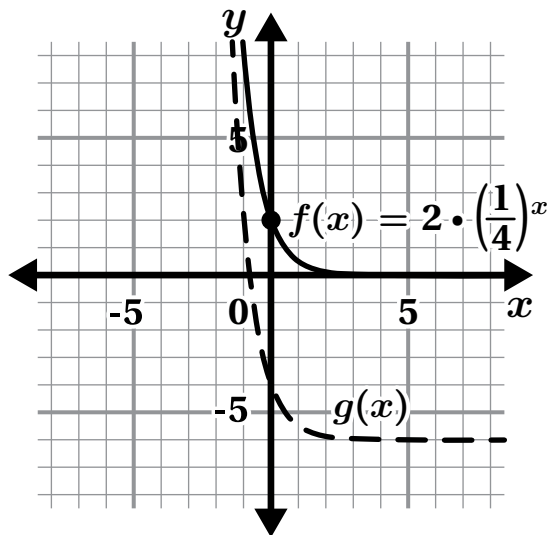
Problems 2–3: For each set of graphs shown, write an equation for the dotted curve, $g(x)$.

2. $f(x) = 3 \cdot \left(\frac{1}{2}\right)^x$

3. $f(x) = 2 \cdot \left(\frac{1}{4}\right)^x$



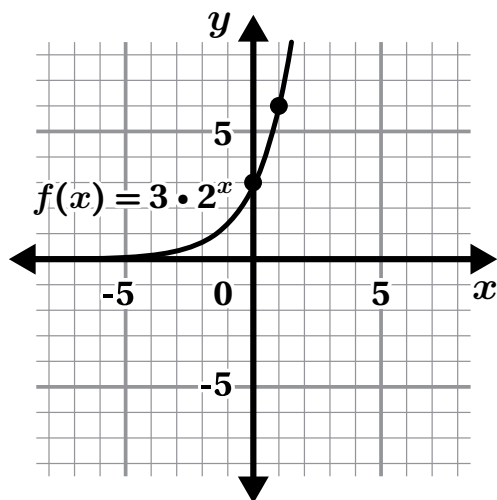
$g(x) = \dots\dots\dots$



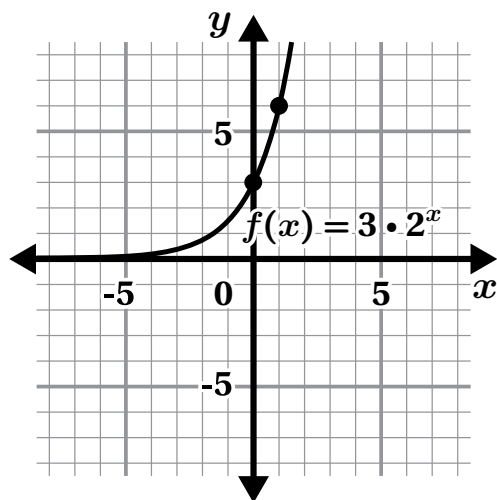
$g(x) = \dots\dots\dots$

Problems 4–5: The function $g(x)$ is a transformation of $f(x) = 3 \cdot 2^x$.

4. Graph $g(x) = 3 \cdot 2^x - 3$

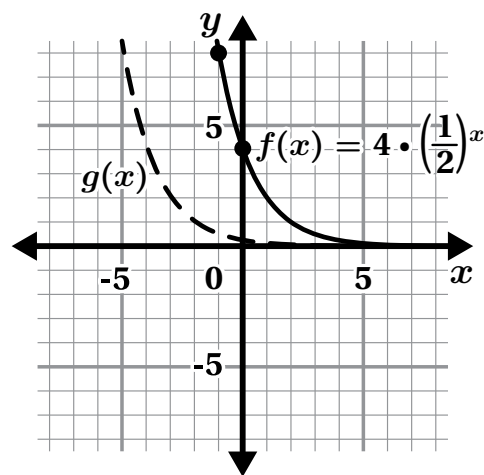


5. Graph $g(x) = 3 \cdot 2^{(x+4)}$



Problems 6–8: The function $g(x)$ is a transformation of $f(x) = 4 \cdot \left(\frac{1}{2}\right)^x$.

6. Melanie says $g(x) = 4 \cdot \left(\frac{1}{2}\right)^{x-4}$ because the point $(0, 4)$ on $f(x)$ shifted to the left 4 units. Explain why Melanie's thinking is incorrect.



7. Yousef says $f(x)$ transformed using a vertical translation because the y -intercept shifted down about 3.5 units. Explain why Yousef's thinking is incorrect.

8. Write the correct equation for $g(x)$.

Additional Practice

A1.6.13

Problems 1–4: Determine if each equation or table represents simple or compound interest. Circle your choice.

1. $b(t) = 500 + 25t$

Simple

Compound

2. $b(t) = 500(1.05)^t$

Simple

Compound

3.

Time (yr)	Account Balance (\$)
0	100
1	125
2	156.25

Simple

Compound

4.

Time (yr)	Account Balance (\$)
0	100
1	125
2	150

Simple

Compound

Problems 5–7: Helen invests \$2,000 in an account that earns 3.5% compound interest per year.

5. Complete the table.

6. Which function represents the amount of money in Helen's account after x years?

A. $f(x) = 2000 + 1.035x$

B. $f(x) = 2000(1.035x)$

C. $f(x) = 2000 + 1.035^x$

D. $f(x) = 2000(1.035)^x$

Time (yr)	Account Balance (\$)
0	
1	2070
2	2142.45
3	
4	

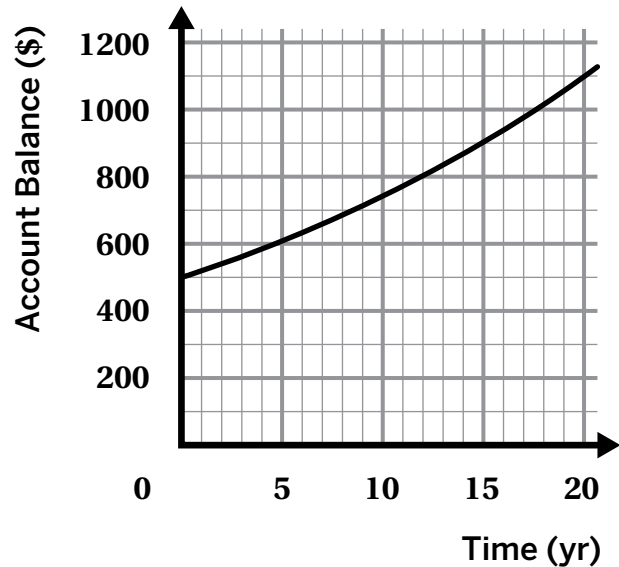
7. What will the balance of the account be after 8 years?

Problems 8–9: Samuel invests \$500 in an account that earns 4% compound interest per year. The graph shows the function $f(t) = 500(1.04)^t$, which gives Samuel's account balance after t years.

8. About how many years will it take for his account balance to reach \$1,000?

9. Use the graph to determine the value of $f(12)$.

What does that tell you about the situation?



10. You just won a contest and have two prize options:

Option A: Getting a penny that doubles every day for 30 days

Option B: One payment of \$10 million dollars

Which option would you choose? Explain your choice.

Additional Practice

A1.6.14

Problems 1–2: Sree takes out a \$2,500 loan with a monthly interest rate of 4.5%. She makes no additional payments, deposits, or withdrawals.

1. Select *all* the expressions that can be used to calculate her balance after t years.

- ☐ A. $2500 \cdot 1.045^t$
- ☐ B. $2500 \cdot (1.045^{12})^t$
- ☐ C. $2500 \cdot 1.045^{12t}$
- ☐ D. $2500(1.6959)$
- ☐ E. $2500 \cdot 1.6959^t$

2. What is the interest rate per year for this loan?

Problems 3–7: Aki invests money into a savings account to purchase a car in the future. He writes the expression $2000(1.055^{12})^4$ to help him calculate what the account balance will be in 4 years.

3. Explain what 2000 represents in the expression.

4. Explain what 1.055 represents in the expression.

5. Explain what 12 represents in the expression.

6. Explain what 4 represents in the expression.

7. Write an equivalent expression that could represent Aki's account balance in 4 years.

Problems 8–9: Jaxson is considering taking out a credit card loan that has a 24% monthly interest rate.

8. Complete the table.

Monthly Interest Rate	24%
Monthly Growth Factor	
Growth Factor Per Year	
Interest Rate Per Year	

- 9.** If Jaxson takes out a \$100 credit card loan, how much will he owe after 3 years, if he made no additional payments?
- 10.** Varsha takes out a \$10,000 federal student loan with a monthly interest rate of 0.64%. Write a function, $f(t)$, to calculate the amount Varsha owes after making no payments for t years.

Additional Practice

A1.6.15

1. Meegan put \$3,000 into a savings account with a 3.25% annual interest rate, compounded quarterly. She made no additional payments, deposits, or withdrawals.

Select *all* the expressions that can be used to calculate her balance after 5 years.

☐ A. $3000(1 + 0.0325)^5$

☐ B. $3000(1 + 0.0325)^{20}$

☐ C. $3000\left(1 + \frac{0.0325}{4}\right)^{20}$

☐ D. $3000(1.008)^{4 \cdot 5}$

☐ E. $3000\left(1 + \frac{0.0325}{4}\right)^4$

Problems 2–4: Amir wants to take out a \$1,200 loan to pay for a new bike. The bank offers him the loan with a 12.5% annual interest rate, compounded daily.

Amir wrote this expression to calculate the balance of the loan in 3 years, but he made an error.

$$1200\left(1 + \frac{0.125}{365}\right)^3$$

2. Find the error and explain why it is incorrect.
3. Write a correct expression to represent Amir's balance after 3 years.
4. What will his balance be in 3 years?

Problems 5–6: A credit card company offers a \$2,500 loan with a 28.8% annual interest rate.

5. If no other charges or payments are made, what will the balance of the loan be after 1 year at each compounding period?

Compounding Period	Balance (\$)
Annually	
Semiannually	
Quarterly	

6. Describe how changing the compounding period affects the balance of the loan.

Problems 7–8: Hannah has \$3,000 to invest and is choosing between three investment options:

- **Option A:** 11.5% annual interest rate, compounded quarterly
- **Option B:** 11.25% annual interest rate, compounded monthly
- **Option C:** 10.5% annual interest rate, compounded daily

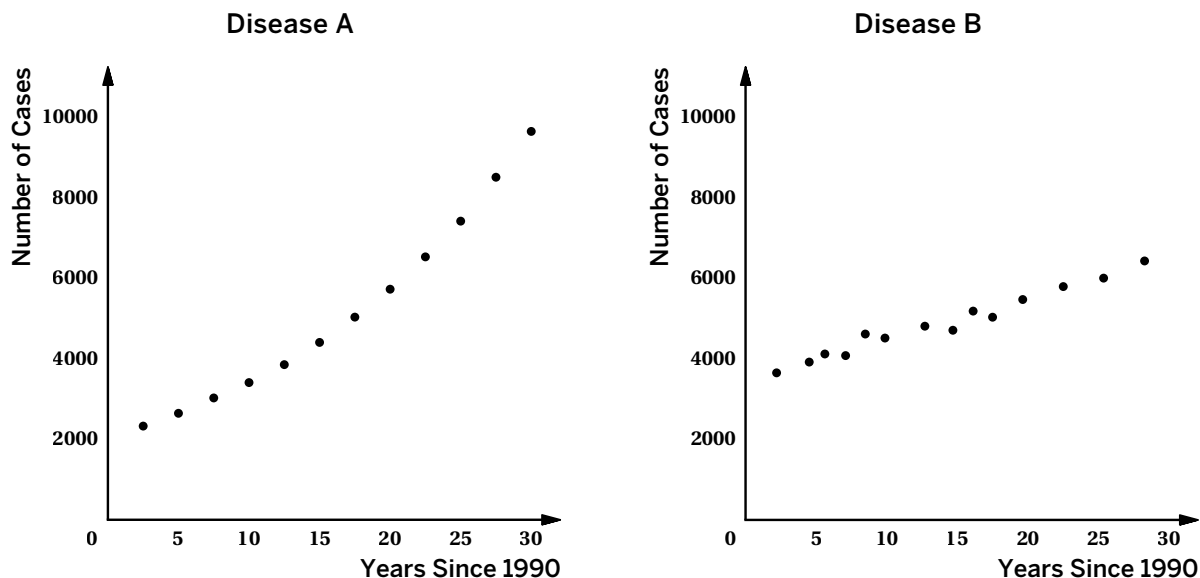
7. Write an expression for each account to represent Hannah's balance after t years.

8. If she makes no deposits and no withdrawals for 6 years, which option will give her the largest balance after 6 years? Show or explain your thinking.

Additional Practice

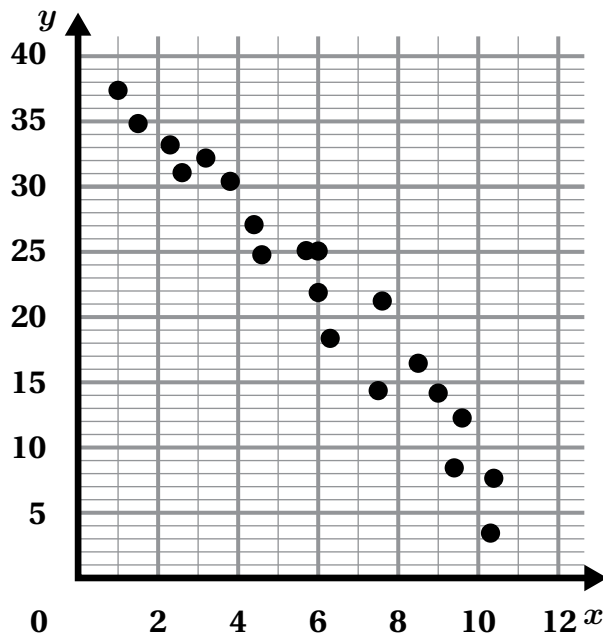
A1.6.16

1. Each graph models the spread of a disease.



What kind of function would best model the data for each disease — *linear*, *exponential*, or *neither*? Explain your thinking.

Problems 2–3: The graph shows data Isabella collected for a research project.



2. Sketch a line or exponential curve of best fit to model the data.

3. Isabella believes that an exponential function is the best model for the data. Do you agree with Isabella? Circle one.

Yes

No

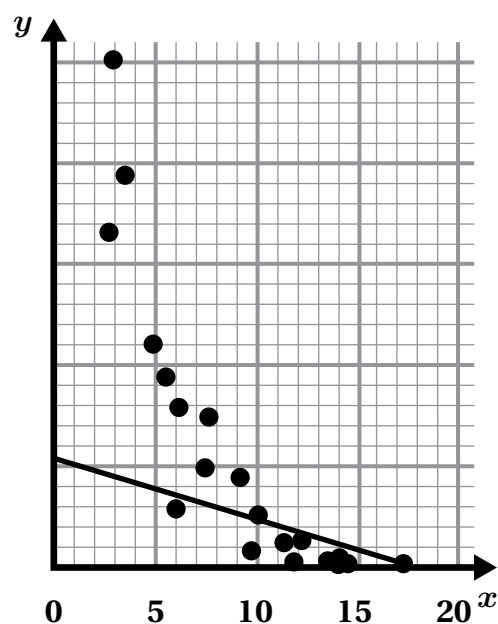
Explain your thinking.

Problems 4–7: The graph shows data from a project in Theo's class. Theo modeled the given data with a linear function.

4. Describe an advantage of using Theo's model.

5. Describe a disadvantage of using Theo's model.

6. Provide a suggestion for Theo to improve his model.



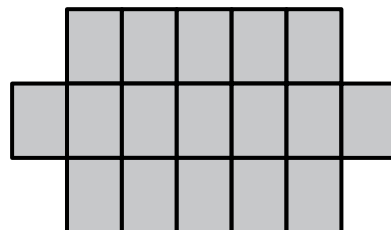
7. Sketch a line or exponential curve of best fit to model the data. Explain your thinking.

Additional Practice

A1.7.01

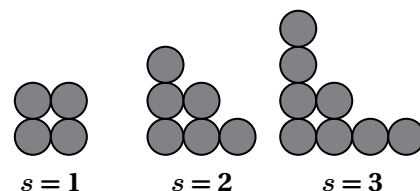
1. Select *all* of the expressions that could represent the number of tiles in this diagram.

- ☐ A. $7 \cdot 3$
- ☐ B. $7 \cdot 3 - 4$
- ☐ C. $5 \cdot 3 + 4$
- ☐ D. $5 \cdot 3 + 2$
- ☐ E. $2(5) + 1(7)$



2. Here are the first three steps in a pattern. How many dots will there be when $s = 8$?

Explain your thinking.



3. What type of relationship does the pattern in the table represent? Circle one.

Linear Exponential Neither

Explain your thinking.

s	Number of Tiles
1	5
2	20
3	45
4	80

Problems 4–6: A teacher gives the class a table with only the first two rows in a tile pattern.

4. Momo says the pattern is a linear relationship.
Gerald says there is not enough information to be sure.
Whose thinking is correct? Explain your thinking.

s	Number of Tiles
1	1
2	8

5. How many tiles would there be in the next step if the relationship were *linear*?
6. How many tiles would there be in the next step if the relationship were *exponential*?

Problems 7–8: The table shows the relationship between the step number, s , and number of tiles in a pattern.

Heather notices that the number of tiles is increasing in an interesting pattern: The number of tiles increases by 3, then by 5, then by 7, and then by 9.

Aaron notices that the number of tiles in each row is the square of the step number.

7. Based on these observations, is this pattern linear, exponential, or neither? Explain your thinking.

s	Number of Tiles
1	1
2	4
3	9
4	16
5	25

8. Use both Aaron and Heather's observations to predict the next number in this pattern. Is it the same value? Show or explain your thinking.

Additional Practice

A1.7.02

Problems 1–4: Circle whether each expression represents a quadratic relationship, a linear relationship, or neither.

- | | | | |
|------------------|-----------|--------|---------|
| 1. $2 \cdot 3^x$ | Quadratic | Linear | Neither |
| 2. $3n^3$ | Quadratic | Linear | Neither |
| 3. $3n^2 - 2n$ | Quadratic | Linear | Neither |
| 4. $n(5n)$ | Quadratic | Linear | Neither |

5. Does this pattern show a quadratic relationship? Explain your thinking.



$n = 1$



$n = 2$

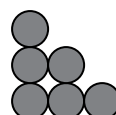


$n = 3$

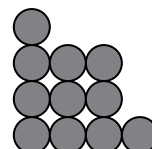
6. Write an expression to represent the relationship between the figure number, n , and the total number of tiles.



$n = 1$

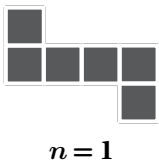


$n = 2$



$n = 3$

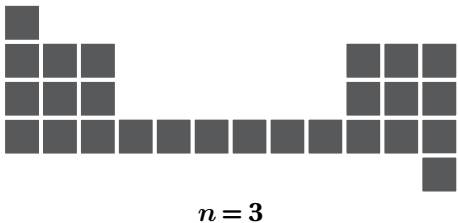
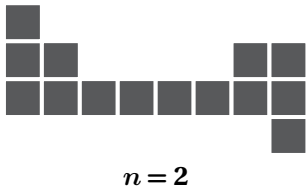
7. Chidi wrote the expression $2n^2 + 2n + 2$ to represent this pattern.
- Denali wrote the expression $2(n^2 + n + 1)$ to represent this pattern.



Whose thinking is correct?

Chidi Denali Both

Explain your thinking



8. Here is a table with the first two rows of a pattern.
- Does this pattern represent a quadratic relationship?
- Yes No Not enough information

s	Number of Tiles
1	10
2	17

Explain your thinking.

Additional Practice

A1.7.04

Problems 1–3: For each pair of symmetrical points on a parabola, determine the equation for the line of symmetry.

1. $(0, 0)$ and $(-12, 0)$

$x =$ _____

2. $(5, 8)$ and $(21, 8)$

$x =$ _____

3. $(10, -3)$ and $(-13, -3)$

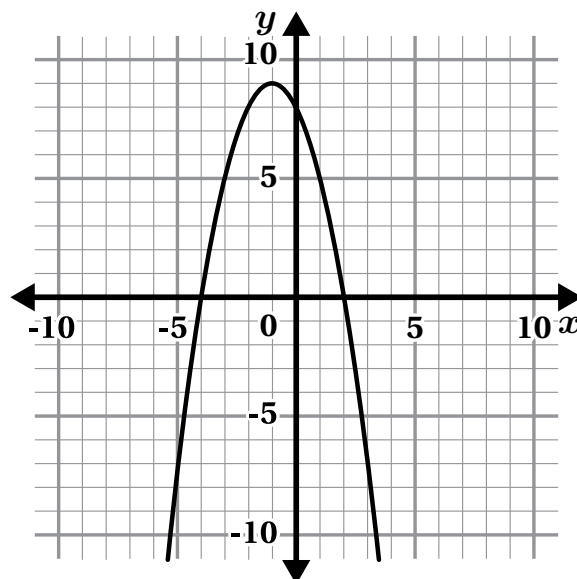
$x =$ _____

Problems 4–5: Here is a graph of a parabola.

4. Draw the *line of symmetry* where you think it is located on this parabola.

5. Write the equation for the line of symmetry.

$x =$ _____



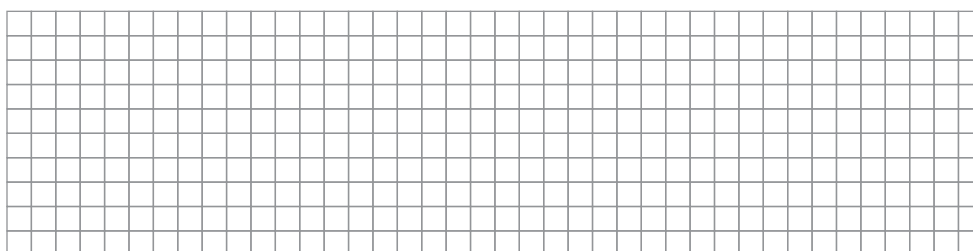
Problems 6–7: Here is an incomplete table that could represent several types of functions.

6. Circle a function type and determine the number of tiles that would be in figure 2.

Linear Quadratic Exponential

Figure	Number of Tiles
1	4
2	
3	16

7. Draw three figures to match the pattern in the table.



Additional Practice

A1.7.05

Problems 1–3: This table shows the height of a stomp rocket at various times.

Time (sec)	0	1	2	3	4	5	6
Height (ft)	8	28	40	44	40	28	8

1. What is the maximum height the stomp rocket reaches?

2. What is the height of the rocket at launch?

3. Issa says the stomp rocket reaches the ground at 6 seconds.

Samnang says the stomp rocket reaches the ground at 7 seconds.

Whose thinking is correct?

Issa Samnang Neither

Explain your thinking.

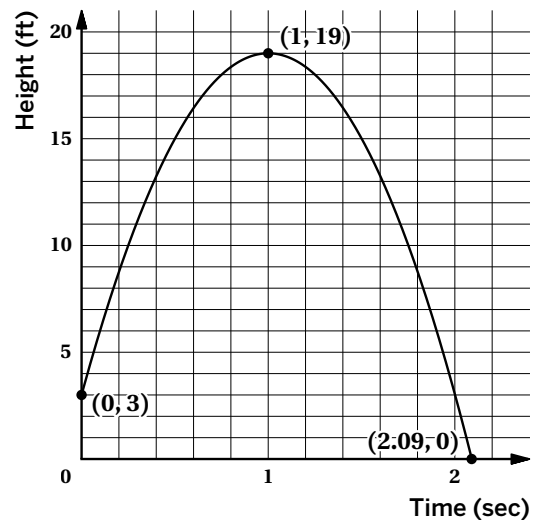
4. $f(x)$ is a quadratic function. Complete the table.

x	0	1	2	3	4	5	6	7
$f(x)$	108	90	76	66				

Describe the strategy you used to complete the table.

Problems 5–7: Shawn throws a ball up in the air. The graph shows the height of the ball, in feet, above the ground as a function of time, in seconds.

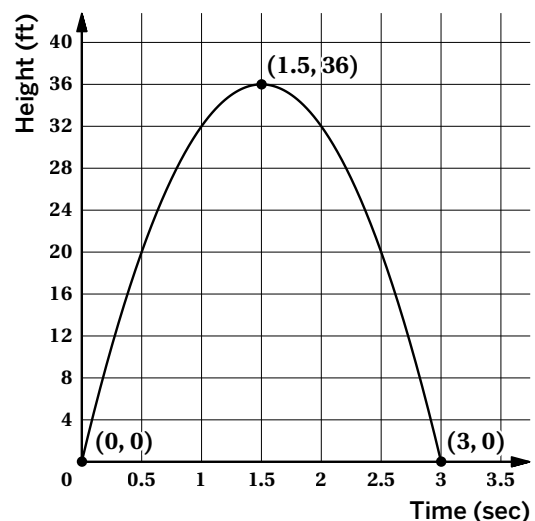
5. How high above the ground was the ball when Shawn threw it?
6. When did the ball reach its maximum height? How high was it?



7. When did the ball hit the ground?

8. A disc is launched into the air. The graph shows the height of the disc, in feet, above the ground as a function of time, in seconds. Select *all* the true statements about the situation.

- ☐ A. The disc was launched from the ground.
- ☐ B. The disc was launched 4 feet from the ground.
- ☐ C. The maximum height of the disc was 36 feet.
- ☐ D. The disc was in the air for 3 seconds.
- ☐ E. The disc was in the air for 36 seconds.
- ☐ F. The disc reached its maximum height 1.5 seconds after it was launched.



Additional Practice

A1.7.06

1. The key features of this parabola are labeled A , B , C , and D .

Match each key feature with a term from the word bank.

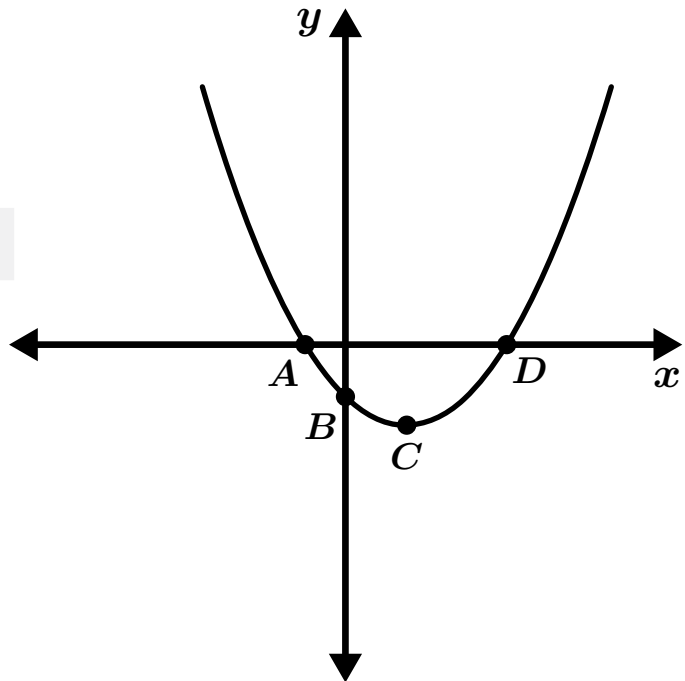
vertex x -intercept y -intercept

A: _____

B: _____

C: _____

D: _____

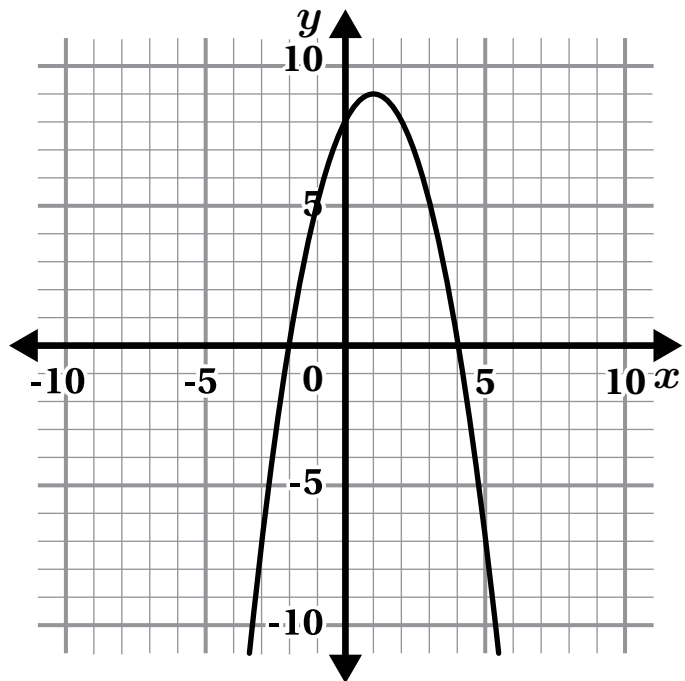


Problems 2–4: Use the graph to determine the coordinates of each key feature.

2. Vertex:

3. x -intercept(s):

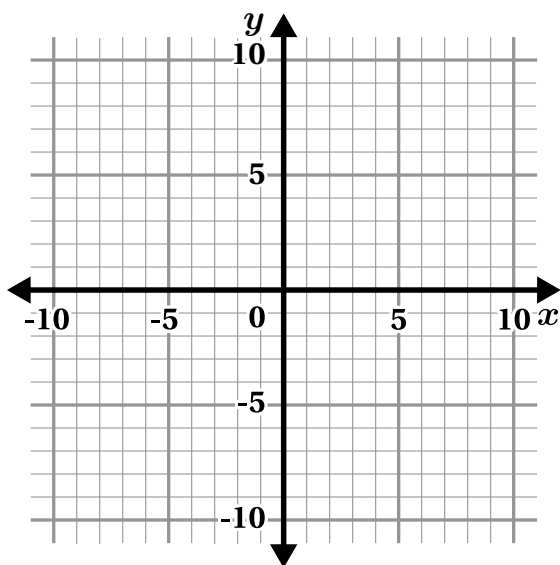
4. y -intercept:



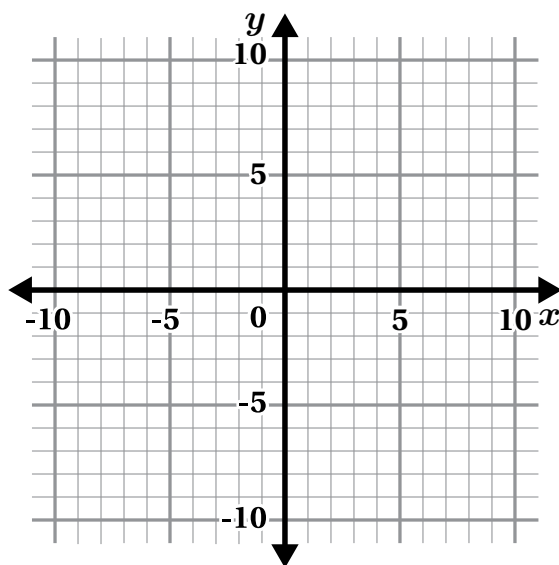
5. A parabola has a vertex at $(-6, -2)$. Give two possible coordinates for its x -intercepts.

Problems 6–9: Graph a parabola that fits each description.

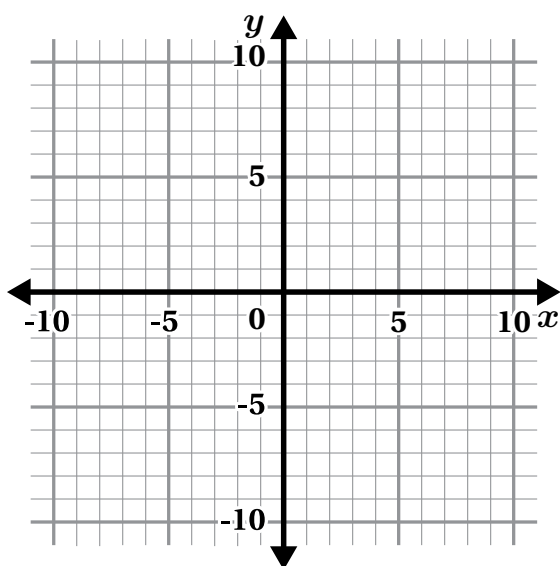
- 6.** Concave up with a negative y -intercept



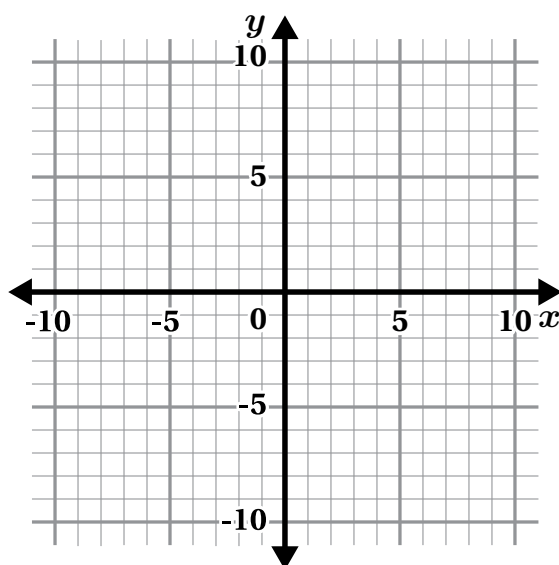
- 7.** Concave down and vertex at $(-3, 4)$



- 8.** Concave down and line of symmetry of $x = 2$



- 9.** Concave up with no x -intercepts

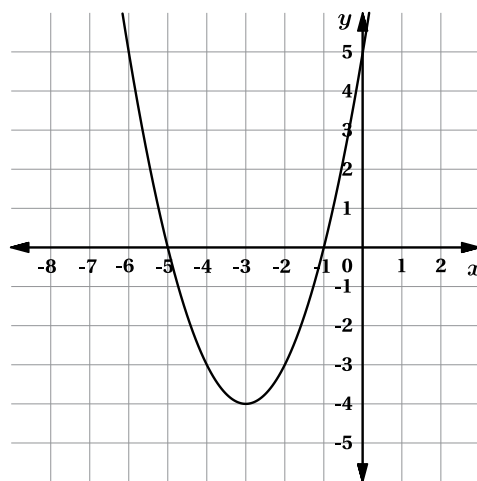


Additional Practice

A1.7.10

1. Here is the graph of $k(x) = (x + 1)(x + 5)$ is shown.

- a What are the x -intercepts?
- b What is the y -intercept?



2. What is the y -intercept of the function $j(x) = x^2 + 4$?

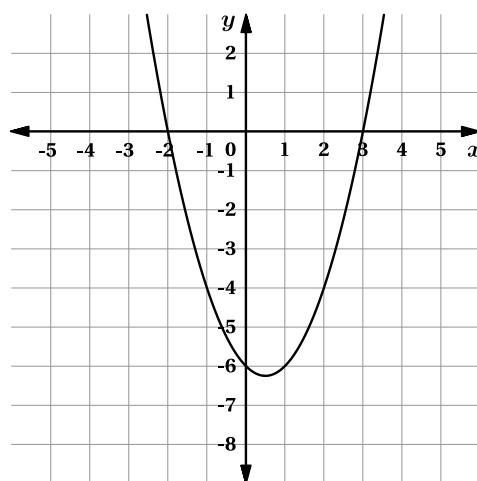
- A. $(0, -4)$
- B. $(0, 0)$
- C. $(0, 1)$
- D. $(0, 4)$

3. What are the x -intercepts of the function $f(x) = (x + 3)(x - 1)$?

- A. $(3, 0)$ and $(1, 0)$
- B. $(-3, 0)$ and $(1, 0)$
- C. $(3, 0)$ and $(-1, 0)$
- D. $(-3, 0)$ and $(-1, 0)$

4. Here is a graph of a quadratic function. Which function could this be?

- A. $g(x) = (x - 2)(x - 3)$
- B. $g(x) = (x - 2)(x + 3)$
- C. $g(x) = (x + 2)(x - 3)$
- D. $g(x) = (x + 2)(x + 3)$



5. Select *all* the quadratic functions that have a y -intercept of $(0, -9)$.

- ☐ A. $f(x) = x(x - 9)$
☐ B. $f(x) = x^2 + 6x + 9$
- ☐ C. $f(x) = x^2 - 5x - 9$
☐ D. $f(x) = (x - 3)(x + 3)$
- ☐ E. $f(x) = (x - 1)(x + 9)$

6. Here is a function: $g(x) = x^2 - 2x - 24$.

- a What is the y -intercept of $g(x)$?
- b Here is the function written a different way: $g(x) = (x - 6)(x + 4)$.
What are the x -intercepts of $g(x)$?

7. Determine the x - and y -intercepts of the function $p(x) = (x - 5)(x + 5)$.

x -intercepts: and

y -intercept:

8. Here is a quadratic function: $r(x) = x(6 - x)$. The table shows where Jada and Elena think the x - and y -intercepts are located. Is either person's work completely correct? Explain your thinking.

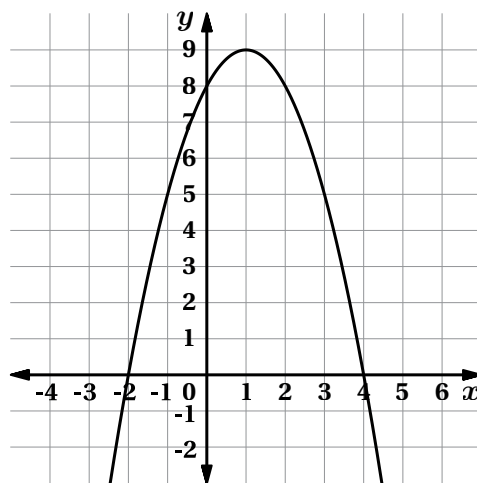
	x -intercepts	y -intercept
Jada	$(-6, 0)$ and $(6, 0)$	$(0, 0)$
Elena	$(0, 0)$ and $(6, 0)$	$(0, -6)$

Additional Practice

A1.7.11

1. Here is the graph of a quadratic function.

- a What is the vertex?
- b What is the equation of the line of symmetry?



2. A quadratic function has a vertex of $(-2, -8)$.

What is the equation of the line of symmetry?

- A. $x = -2$
- B. $x = -8$
- C. $x = 2$
- D. $x = 8$

3. Select *all* the true statements about the function $f(x) = x(x + 10)$.

- ☐ A. The x -coordinate of its vertex is -5 .
- ☐ B. The x -coordinate of its vertex is 5 .
- ☐ C. The x -coordinate of its vertex is 10 .
- ☐ D. It has exactly two x -intercepts.
- ☐ E. The x -intercepts are $(-1, 0)$ and $(-10, 0)$.
- ☐ F. The x -intercepts are $(0, 0)$ and $(-10, 0)$.
- ☐ G. The only x -intercept is $(10, 0)$.

4. Select *all* the functions that have a vertex with an x -coordinate of -3 .

- | | |
|---|---|
| ☐ A. $g(x) = (x - 3)(x + 3)$ | ☐ B. $g(x) = (x + 2)(x + 4)$ |
| ☐ C. $g(x) = (x - 1)(x - 5)$ | ☐ D. $g(x) = x(x + 6)$ |
| ☐ E. $g(x) = x(x - 6)$ | |

5. Determine the x -intercepts and the line of symmetry of the function

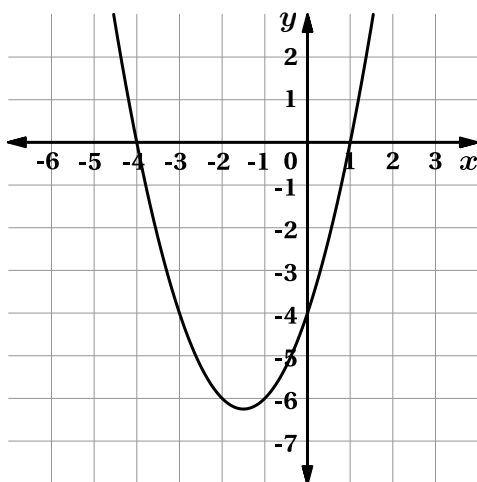
$$j(x) = (x + 6)(x + 2).$$

x -intercepts: and

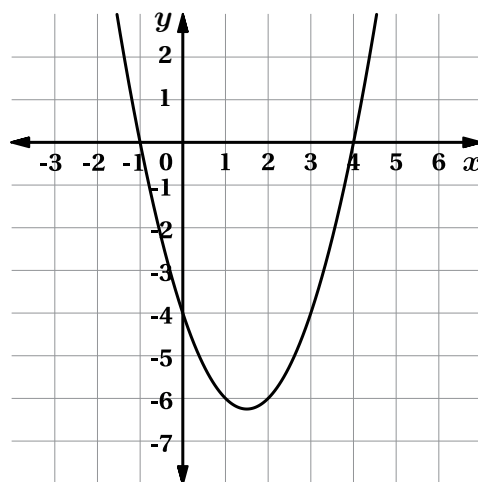
Line of symmetry: $x =$

6. Here are the graphs of two quadratic functions. Which is the graph of $w(x) = (x + 4)(x - 1)$? Explain your thinking.

Graph A



Graph B



7. The quadratic function $r(x)$ has x -intercepts at $(-6, 0)$ and $(1, 0)$. What is the equation for the axis of symmetry?

8. Here are the two quadratic functions:

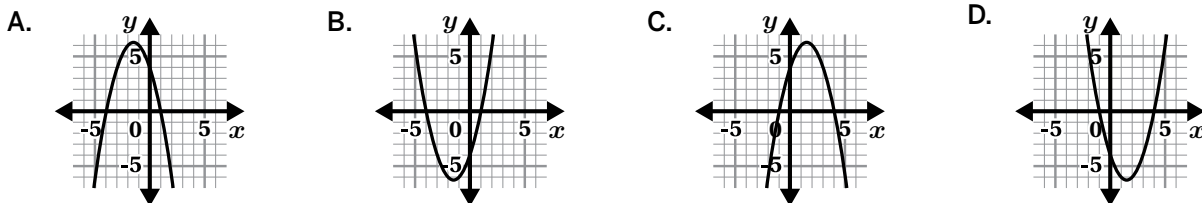
$$f(x) = (x + 3)(x - 11) \qquad g(x) = 2x(x - 8)$$

Kiran claims that the axis of symmetry for both functions is $x = 4$. Do you agree with Kiran's claim? Show or explain your thinking.

Additional Practice

A1.7.12

1. Which graph shows the function $y = -(x - 4)(x + 1)$?



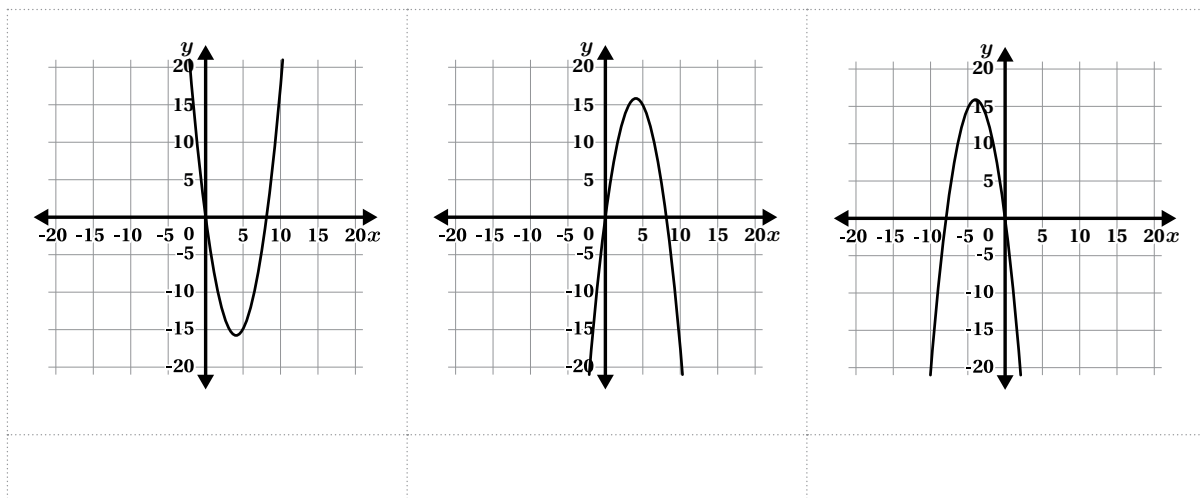
2. Match each equation to the graph it represents. One equation will have no match.

$$y = -x(x - 8)$$

$$y = x(x - 8)$$

$$y = x(x + 8)$$

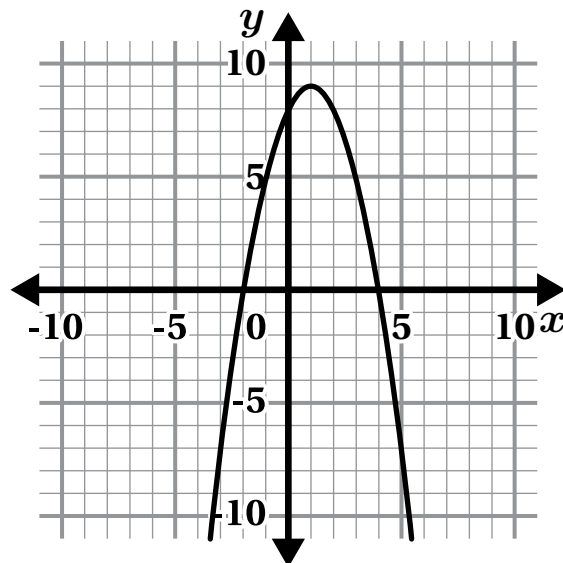
$$y = -x(x + 8)$$



Problems 3–4: Here is a graph of $y = -1(x + 2)(x - 4)$. Change the equation so the vertex goes through:

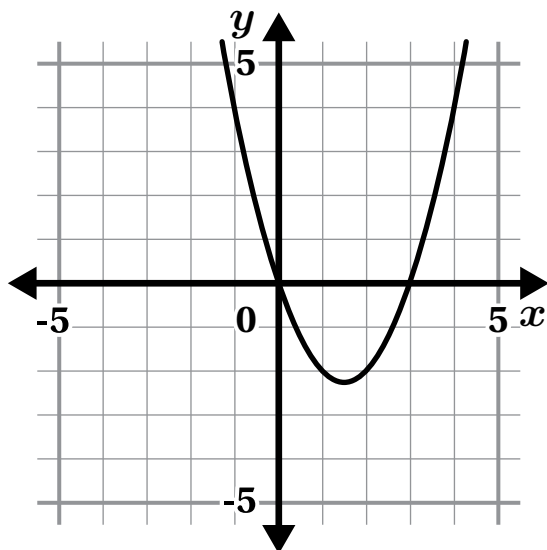
3. $(1, -9)$

4. $(1, 4.5)$

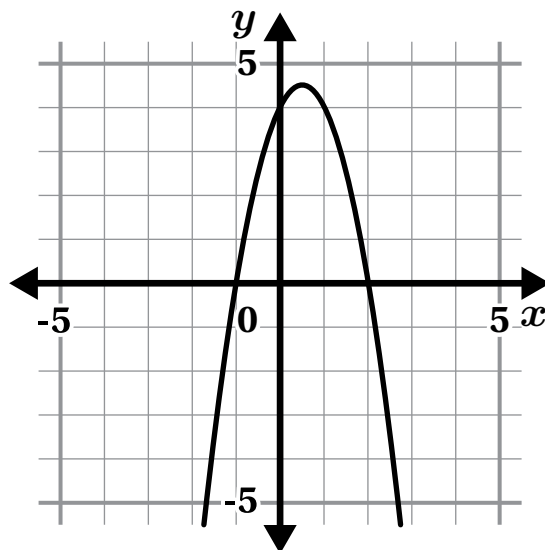


Problems 5–6: Write an equation to match each graph.

5. Equation:



6. Equation:



7. Write the equations of three different quadratic functions that have the same x -intercepts but different y -intercepts.

Equation 1	Equation 2	Equation 3

Additional Practice

A1.7.14

Problems 1–2: Circle if the transformation of $y = x^2$ is a *horizontal translation*, *vertical translation*, or *neither*.

1. $y = x^2 - 7$

Horizontal
translation

Vertical
translation

Neither

2. $y = 6x^2$

Horizontal
translation

Vertical
translation

Neither

3. These parabolas are translations of $y = x^2$.

Select *all* of the equations shown on the graph.

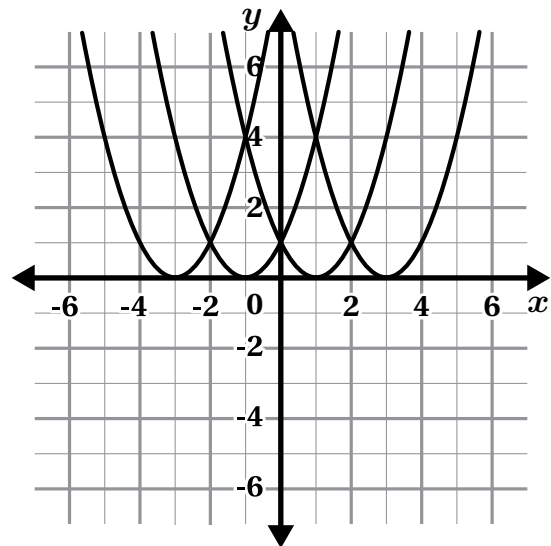
☐ A. $y = x^2 + 3$

☐ B. $y = (x - 3)^2$

☐ C. $y = (x + 3)^2$

☐ D. $y = x^2 - 1$

☐ E. $y = (x + 1)^2$



4. Match each vertex of a parabola to its equation.

a $y = x^2 - 3$ (3, -4)

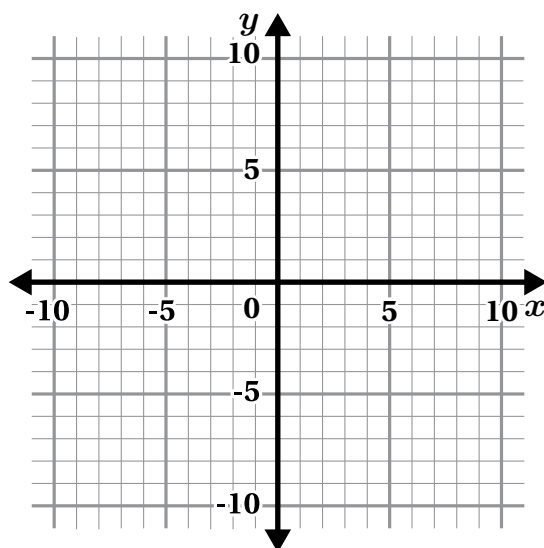
b $y = (x - 3)^2 - 4$ (3, 0)

c $y = (x - 3)^2$ (-3, -4)

d $y = (x + 3)^2 - 4$ (0, -3)

e $y = (x - 4)^2 - 3$ (4, -3)

5. Draw the graph of $y = (x - 2)^2 + 5$.



6. Here are four equations in vertex form. Which function has a graph with a vertex at $(-4, 5)$?

A. $y = (x - 4)^2 + 5$

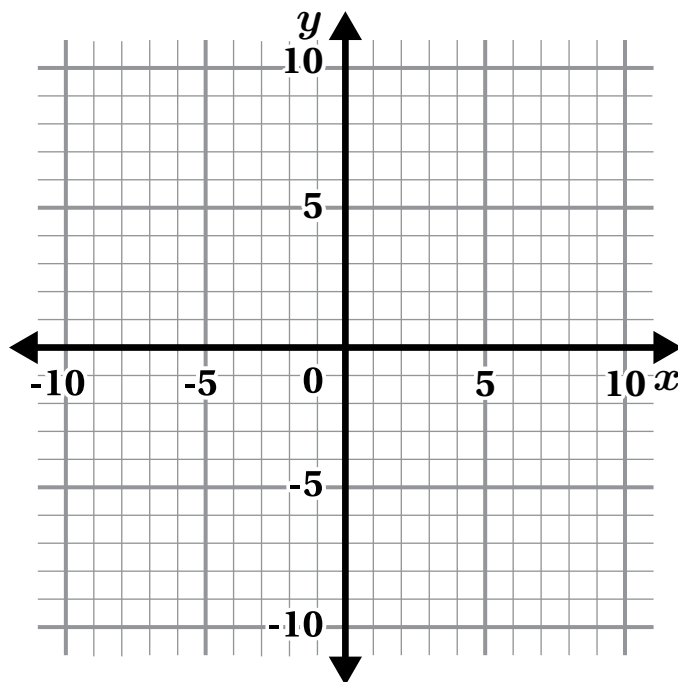
B. $y = (x + 4)^2 - 5$

C. $y = (x - 5)^2 + 4$

D. $y = (x + 4)^2 + 5$

7. Write and graph the equations of five parabolas so that when you connect their vertices, they form the line $y = \frac{1}{2}x$.

Equations



Additional Practice

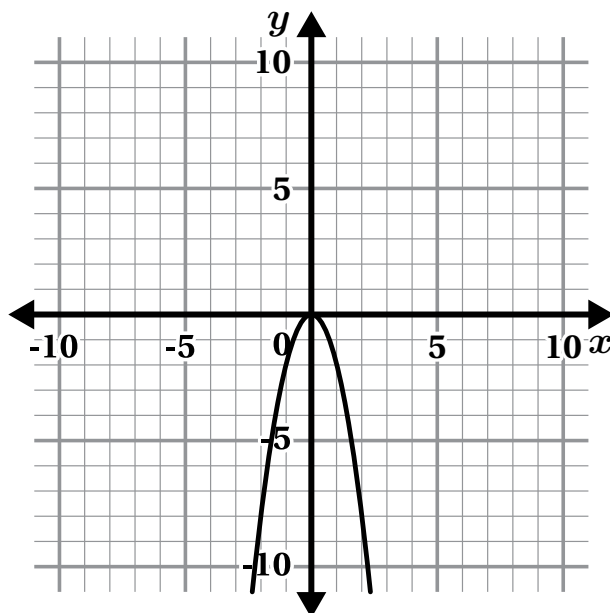
A1.7.15

Problems 1–3: Here's the graph of $y = -2x^2$. Change one number to make the graph:

1. Wider:

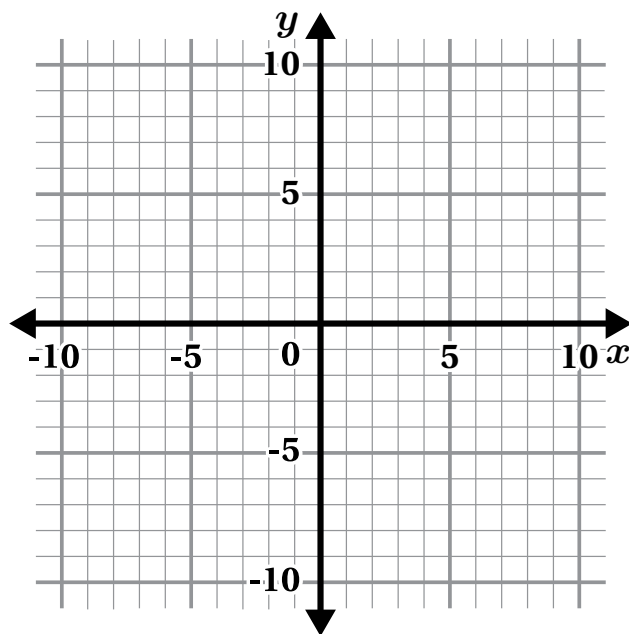
2. Narrower:

3. Concave up:



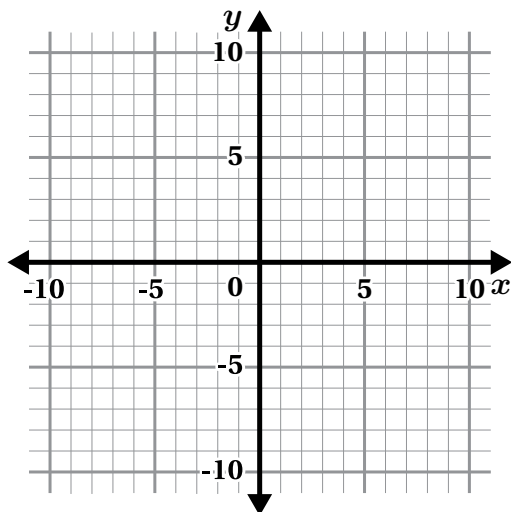
4. Describe how the graph of $f(x) = x^2$ compares to $g(x) = 3(x + 1)^2 - 4$.

5. Draw a graph of a parabola that has a vertex at $(-1, 3)$ and is scaled vertically by a factor of $\frac{1}{2}$.

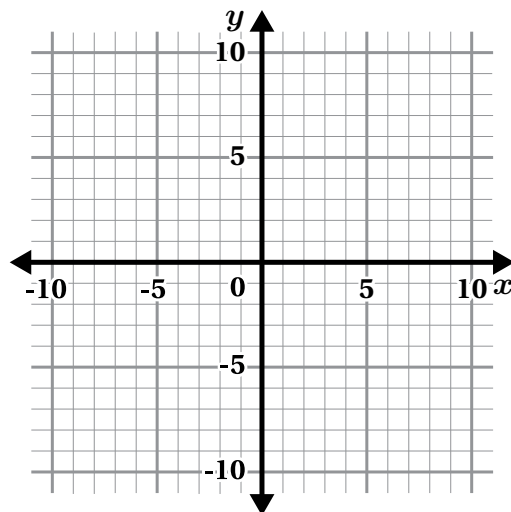


Problems 6–7: Draw a graph for each equation.

6. $c(x) = 0.5(x - 1)^2 + 3$



7. $d(x) = -3(x + 2)^2 + 4$



8. The graph of $h(x) = x^2$ is transformed so that its new equation is $j(x) = -2(x - 9)^2 + 6$. Select *all* of the true statements.

- ☐ A. The graph of $j(x)$ is concave up.
- ☐ B. The graph of $j(x)$ is concave down.
- ☐ C. The graph of $j(x)$ is narrower than $h(x)$.
- ☐ D. The graph of $j(x)$ is wider than $h(x)$.
- ☐ E. The vertex of $h(x)$ was translated 9 units to the right and 6 units up to get to the vertex of $j(x)$.
- ☐ F. The vertex of $h(x)$ was translated 9 units to the left and 6 units down to get to the vertex of $j(x)$.

Additional Practice

A1.7.16

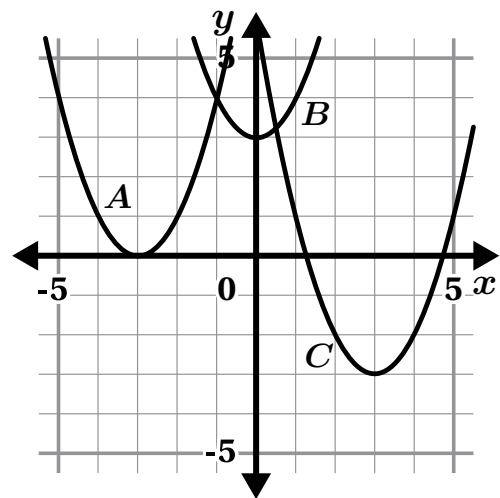
1. Which equation has a graph with a vertex at $(5, 8)$?

- A. $y = (x + 5)^2 + 8$
- B. $y = (x - 5)^2 + 8$
- C. $y = (x + 8)^2 + 5$
- D. $y = (x - 8)^2 + 5$

2. Parabolas A , B , and C are translations of $y = x^2$.

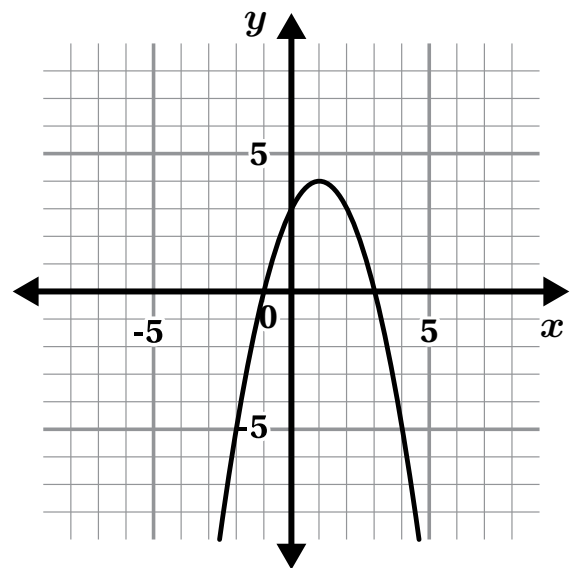
Determine the equation of each parabola.

- A: _____
- B: _____
- C: _____



3. Select *all* the equations that match this graph.

- ☐ A. $y = -(x + 1)^2 - 4$
- ☐ B. $y = -(x + 1)(x - 3)$
- ☐ C. $y = -(x - 1)^2 + 4$
- ☐ D. $y = (x - 1)(x + 3)$
- ☐ E. $y = (x - 1)^2 + 4$



4. Write an equation for a quadratic function with x -intercepts at $(4, 0)$ and $(-7, 0)$.

Problems 5–6: Write an equation for a quadratic function that matches each description. Use graphing technology to check your work.

5. Concave up with a vertex at $(0, 5)$

6. Concave down with a vertex at $(-2, 0)$

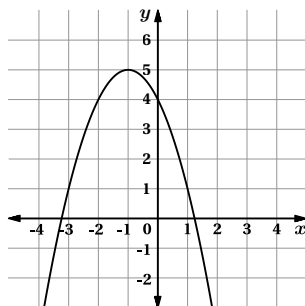
7. Match each graph with the equation that it represents.

..... $y = -(x + 1)^2 + 5$

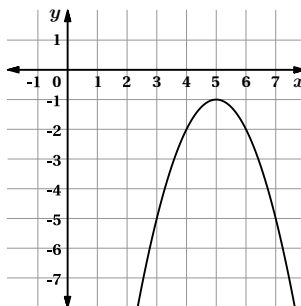
..... $y = (x - 5)^2 + 1$

..... $y = -(x - 5)^2 - 1$

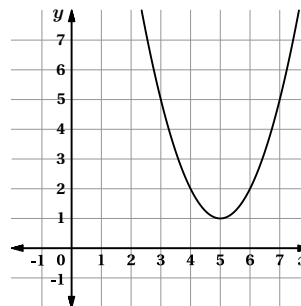
a.



b.



c.

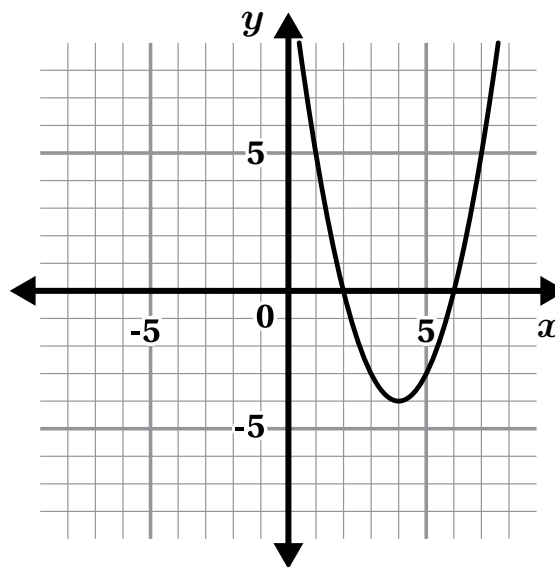


8. Here are two equations:

$$f(x) = (x - 2)(x - 6)$$

$$g(x) = (x - 4)^2 - 4$$

Show or explain how you know that both equations describe this graph.



Additional Practice

A1.7.17

1. Which function has a graph with a vertex at $(-3, 7)$?

A. $y = (x - 3)^2 + 7$

B. $y = (x - 3)^2 - 7$

C. $y = (x + 3)^2 + 7$

D. $y = (x + 3)^2 - 7$

2. Select *all* the functions whose graphs have an x -intercept at $(-2, 0)$.

☐ A. $a(x) = (x - 2)(x - 4)$

☐ B. $b(x) = (x + 2)(x + 4)$

☐ C. $c(x) = 2x(x + 6)$

☐ D. $d(x) = (2x + 4)(x + 3)$

☐ E. $e(x) = (3x - 6)(x + 5)$

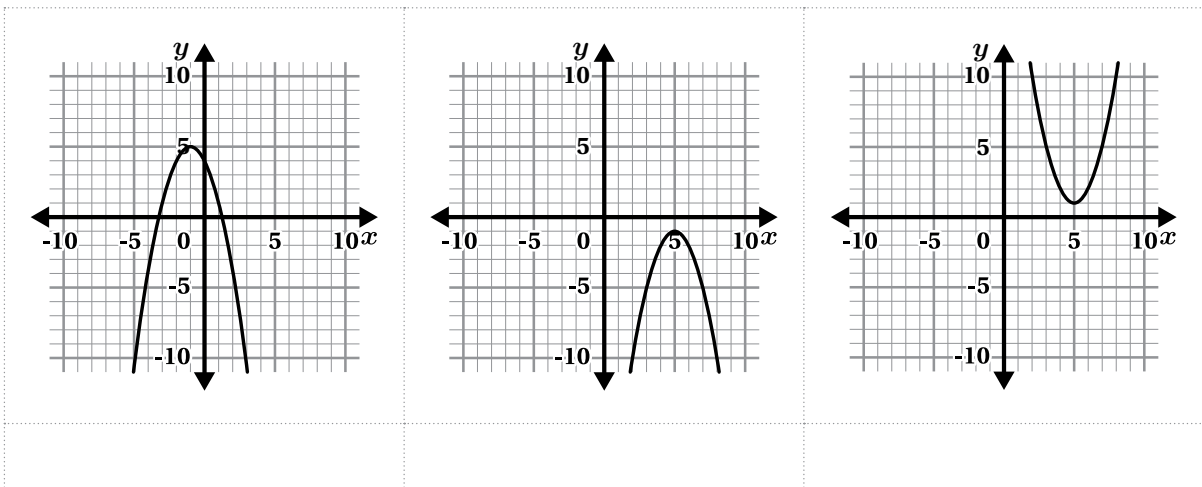
3. Match each equation to the graph it represents. One equation will have no match.

$$y = -(x - 1)^2 + 5$$

$$y = -x^2 - 2x + 4$$

$$y = -(x - 5)^2 - 1$$

$$y = (x - 5)^2 + 1$$



4. Write an equation of a parabola that opens downward and has x -intercepts at $(-4, 0)$ and $(6, 0)$.

Problems 5–6: Terrence and Raj try to write an equation to describe this graph. At least one equation is incorrect.

Terrence

$$y = (x + 2)^2 - 25$$

Raj

$$y = -(x + 3)(x - 7)$$

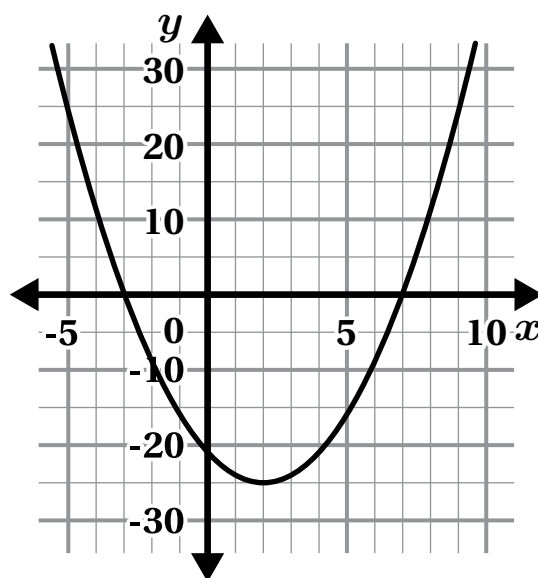
5. Whose equation is correct? Circle one.

Terrence

Raj

Neither

6. Explain how you would change the incorrect equation(s) so that it creates this graph.



Problems 7–8: Here is a function: $r(x) = -x(x + 4)$.

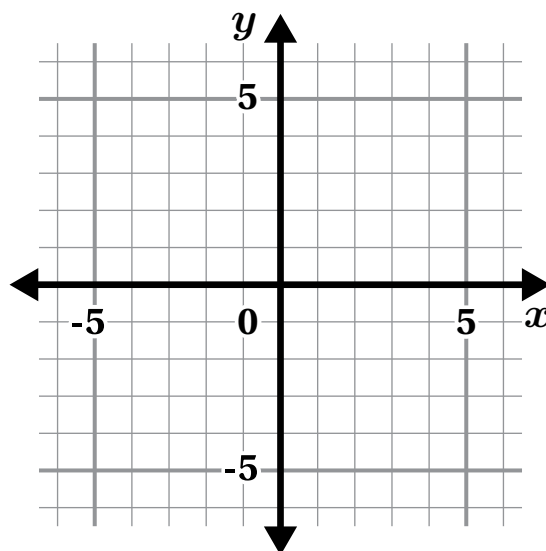
7. Determine the x -intercepts and vertex of $r(x)$.

x -intercept:

x -intercept:

Vertex:

8. Draw the graph of the function $r(x)$.



Additional Practice

A1.8.02

1. Match each expression to its equivalent expression in standard form.

- | | |
|----------------------|-------------------------|
| a. $(2x + 3)(x + 4)$ | $x^2 + 10x + 16$ |
| b. $(x + 3)(x + 2)$ | $2x^2 + 11x + 12$ |
| c. $(x + 8)(x + 2)$ | $x^2 + 5x + 6$ |
| d. $(x + 3)(2x + 1)$ | $2x^2 + 7x + 3$ |

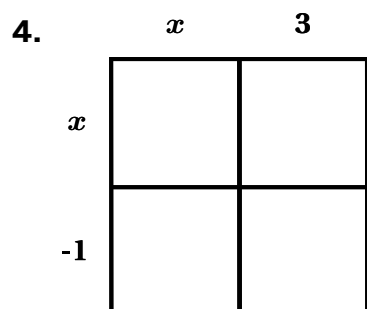
2. Complete the table by writing each expression in standard form.

Factored Form	Standard Form
$(x - 2)(x - 4)$	
$(x + 5)(x + 2)$	

3. Which expression is the standard form of $(x + 1)(2x - 3)$?

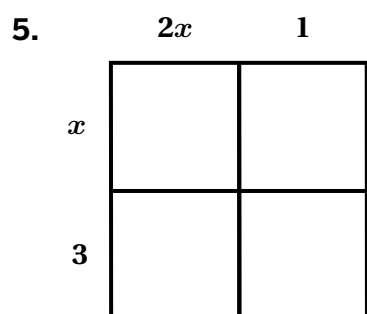
- A. $2x^2 - 2x - 6$
- B. $2x^2 - x - 3$
- C. $3x^2 - 2x - 6$
- D. $3x^2 - x - 3$

Problems 4–6: For each area model, write two expressions, one in factored form and one in standard form.



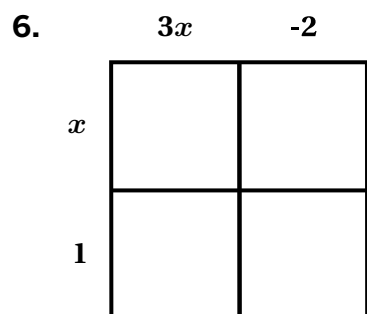
Factored form:

Standard form:



Factored form:

Standard form:



Factored form:

Standard form:

Additional Practice

A1.8.03

1. Is the quadratic expression $2x(x - 1)$ in standard form or in factored form?

2. Which quadratic expression is written in standard form?
 - A. $(x + 3)^2 - 2$
 - B. $(x - 8)3x$
 - C. $4x^2 + 7(x - 1)$
 - D. $-2x^2 + 7x + 4$

3. Consider the quadratic expression $3x^2 - x$.
 - a Is the expression in standard form? Explain your thinking.

 - b Is the expression equivalent to $x(3x - 1)$? Explain your thinking.

4. Which expression is equivalent to $(4x - 2)(2x + 3)$?
 - A. $8x^2 - 4x + 6$
 - B. $8x^2 + 12x + 6$
 - C. $8x^2 + 8x - 6$
 - D. $8x^2 + 10x - 6$

5. Match each quadratic expression in factored form with an equivalent expanded expression in standard form.

Factored Form

Standard Form

- | | |
|----------------------|-------------------------|
| a. $(x + 3)(x + 8)$ | $x^2 - 8x + 12$ |
| b. $(2x + 4)(x - 3)$ | $x^2 + 11x + 24$ |
| c. $(2x + 6)(x + 4)$ | $2x^2 - 2x - 12$ |
| d. $(x - 2)(x - 6)$ | $2x^2 + 14x + 24$ |

- 6.** For each expression in factored form, write an equivalent expression in standard form. Draw a diagram to show your thinking.

a $(x - 2)(x + 7)$

b $(3x + 1)(x - 1)$

c $(x - 6)^2$

d $(2x + 4)(3x + 5)$

- 7.** Here are two quadratic expressions:

$$(x + 3)^2 - 1$$

$$(x + 4)(x + 2)$$

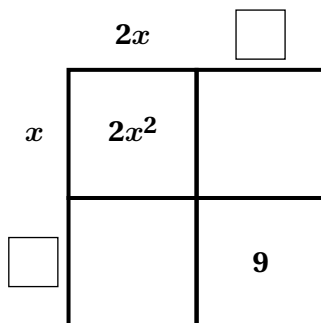
Are the expressions equivalent? Explain your thinking.

- 8.** Here is the quadratic expression $(2x - 6)(-x + 5)$. Mai claims that when the expression is written in standard form, it has two negative terms. Han claims that when the expression is written in standard form, it has three negative terms. Who is correct? Show or explain your thinking.

Additional Practice

A1.8.04

1. Write two possible constants that could complete the outside of the diagram.



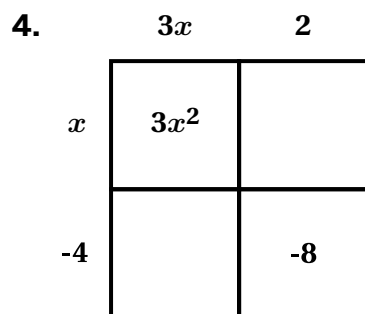
2. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$(3x + 1)(3x - 1)$	
	$4x^2 - 7x - 2$
$(2x + 3)(x - 6)$	
	$x^2 + 6x + 9$

3. Determine values that make the equation true.

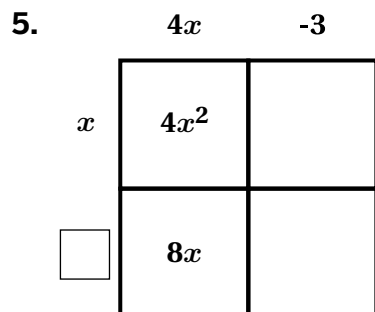
$$2x^2 + x - \underline{\hspace{1cm}} = (\underline{\hspace{1cm}}x - 1)(2x + \underline{\hspace{1cm}})$$

Problems 4–5: Complete the diagram puzzles and expressions.



Factored form:

Standard form:



Factored form:

Standard form:

- 6.** This quadratic expression in standard form has an unknown c -value. Given that we know the expression can be factored, fill in the c -value.

$$3x^2 - 7x - \dots$$

Additional Practice

A1.8.05

1. Select *all* expressions that are equivalent to the expression $x + (-7)$.

☐ A. $7 + x$

☐ B. $x - 7$

☐ C. $7 - x$

☐ D. $-7 + x$

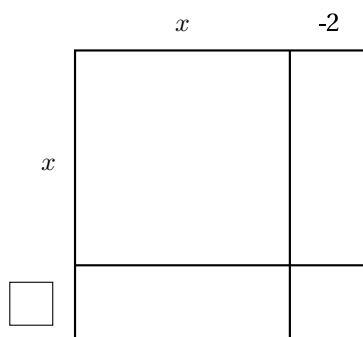
☐ E. $7 + (-x)$

☐ F. $-7 - (-x)$

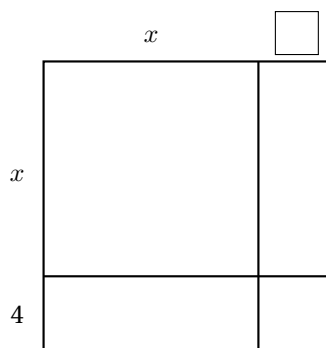
☐ G. $-x - (-7)$

2. Complete the diagrams to show that each pair of expressions is equivalent.

a $(x - 2)(x - 5)$ and $x^2 - 7x + 10$

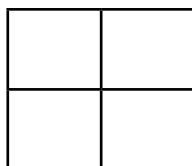


b $(x + 9)(x + 4)$ and $x^2 + 13x + 36$

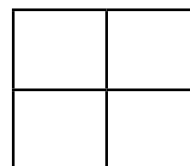


3. Fill in the blanks to make each equation true. Use the diagrams if they help your thinking.

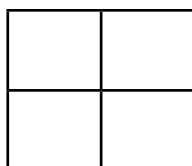
a $x^2 \dots x \dots = (x + 8)(x + 1)$



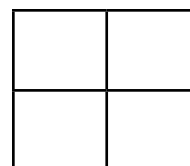
b $x^2 + 5x + 6 = (x + 2)(x \dots)$



c $x^2 - 16x + 63 = (x \dots)(x - 7)$



d $x^2 - 9x + 20 = (x - 5)(x \dots)$



4. Rewrite each quadratic expression in standard form.

a $(x + 6)(x + 8)$

b $(x - 1)(x - 3)$

5. Factor each expression.

a $x^2 - 12x + 20$

b $x^2 + 14x + 49$

c $x^2 - 17x + 72$

d $x^2 + 9x + 20$

6. Rewrite each expression in factored or standard form.

Factored Form	Standard Form
$x(x + 4)$	
	$x^2 - 7x$
	$x^2 + 13x + 22$
$(x - 5)(x - 5)$	
	$x^2 - 12x + 27$

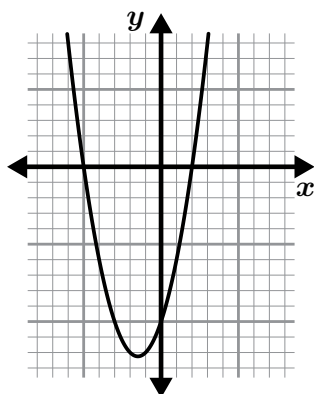
7. Here are two expressions: $(x + n)(x + n)$ and $x^2 + nx + n^2$. Tyler claims the expressions are equivalent. Shawn claims the expressions are not equivalent. Who is correct? Explain your thinking.

Additional Practice

A1.8.06

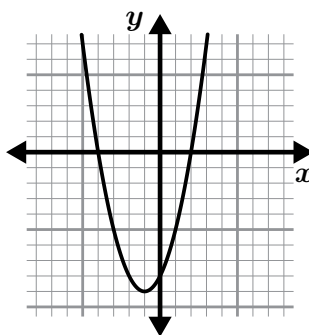
1. What are the x -intercepts of the function $f(x) = (x + 5)(x - 2)$?

- A. $(-5, 0)$ and $(2, 0)$
- B. $(5, 0)$ and $(-2, 0)$
- C. $(0, 5)$ and $(0, 2)$
- D. $(2, 5)$ and $(0, 0)$



Problems 2–3: What are the x -intercepts of the function?

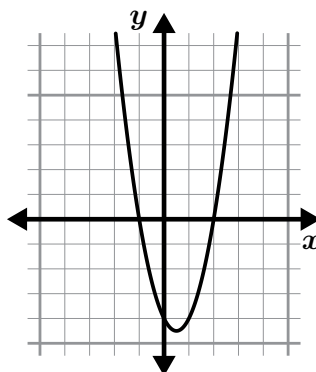
2. $f(x) = x^2 + 2x - 8$



x -intercept 1:

x -intercept 2:

3. $g(x) = 2x^2 - 2x - 4$



x -intercept 1:

x -intercept 2:

4. Select *all* the functions that have $(3, 0)$ and $(-2, 0)$ as their x -intercepts.

☐ **A.** $f(x) = (x + 3)(x - 2)$

☐ **B.** $g(x) = (x - 3)(x + 2)$

☐ **C.** $h(x) = x^2 - x - 6$

☐ **D.** $j(x) = 2x^2 - x - 6$

☐ **E.** $k(x) = (2x + 3)(2x - 2)$

5. How many x -intercepts does the function $g(x) = x^2 + 6x + 9$ have?

A. 0

B. 1

C. 2

D. There is not enough information.

Explain your thinking.

Additional Practice

A1.8.07

1. Bard is solving the equation $x^2 - 8x = 20$. Bard wants to rewrite the equation so that one side is equal to 0. What should be Bard's first step?

- A. Factor $x^2 - 8x$.
- B. Divide each side by x .
- C. Add 20 to both sides of the equation.
- D. Subtract 20 from both sides of the equation.

2. Rewrite each standard form quadratic equation in factored form.

a $x^2 + 2x - 8$

b $x^2 - 2x - 15$

3. Solve each equation.

a $n(n + 9) = 0$

b $(6 + y)(6 - y) = 0$

c $(3x - 1)(x + 11) = 0$

d $(4p + 8)(4p + 8) = 0$

e $(b + 7)(10 - b) = 0$

4. Solve each equation.

a $x^2 - 11x + 10 = 0$

b $t^2 + 24t + 144 = 0$

c $c^2 - c - 72 = 0$

5. $x = 3$ is one of the solutions to the equation $x^2 - x - 6 = 0$. What is the other solution?

6. Write two equations with $x = -7$ and $x = \frac{1}{2}$ as solutions.

Equation 1:

Equation 2:

7. Lin is solving the quadratic equation $x^2 + 8x - 48 = 0$. Her work is shown. Is Lin's work correct or incorrect? If it is incorrect, explain the error and correct it. If it is correct, check Lin's solutions by substituting them into the original equation and showing that the equation is true.

Lin

$$x^2 + 8x - 48 = 0$$

$$(x - 12)(x + 4) = 0$$

$$x - 12 = 0 \text{ or } x + 4 = 0$$

$$x = 12 \text{ or } x = -4$$

8. A square pool has a walkway surrounding it. The total area of the pool and walkway is given by the equation $y = (x + 4)(x + 6)$, where y represents the area in square feet and x represents the side length of the pool in feet. The total area is 360 square feet.
- a Write an equation to represent the total area of the pool and walkway.
- b What is the side length of the pool? Show or explain your thinking.

Additional Practice

A1.8.08

1. How many solutions does the equation have? $(x + 10)^2 = -25$

- A. No Solutions
- B. One Solution
- C. Two Solutions
- D. There is not enough information.

2. For each equation, determine the number of solutions.

Equation	Number of Solutions
$x^2 + 2 = 2$	
$(x - 3)^2 = 1$	
$(x + 8)^2 = -16$	
$(x + 3)^2 = 0$	
$x^2 - 8 = -4$	

3. Which value of x is a solution to the equation $(x - 2)^2 + 27 = 27$?

- A. $x = -2$
- B. $x = 0$
- C. $x = 2$
- D. There is no solution to the equation.

Problems 4–5: Determine the solution for each equation.

4. $(x + 2)^2 = 0$

5. $(x + 3)(x + 3) = 0$

Problems 6–7: Determine the two solutions for each equation. Show your thinking.

6. $x^2 - 20 = -4$

7. $100 + (x - 3)^2 = 136$

8. Which value for x is a solution to the equation $x^2 + 7x = x - 9$?

A. $x = -9$

B. $x = -3$

C. $x = 0$

D. $x = 3$

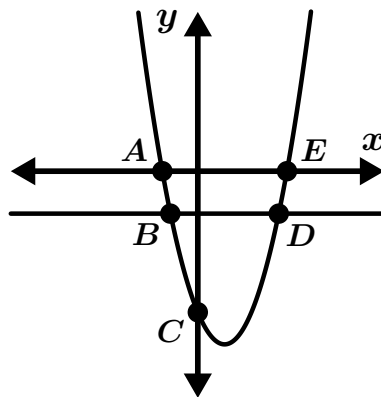
Explain your thinking.

Additional Practice

A1.8.09

1. Here is a graph of $y = x^2 - 3x - 10$ and $y = -4$.
Select *all* the points that are solutions to $x^2 - 3x - 10 = -4$.

- ☐ A. Point A
☐ B. Point B
☐ C. Point C
☐ D. Point D
☐ E. Point E



Problems 2–5: Circle how many solutions each equation has. Record any solutions.

2. $3x^2 + 6 = 18$ No solutions One solution Two solutions $x = \dots\dots\dots$
 $x = \dots\dots\dots$

3. $x(x - 5) = -6$ No solutions One solution Two solutions $x = \dots\dots\dots$
 $x = \dots\dots\dots$

4. $0 = -(x - 3)(x + 2) - 10$ No solutions One solution Two solutions $x = \dots\dots\dots$
 $x = \dots\dots\dots$

5. $x(x + 4) = -16$ No solutions One solution Two solutions $x = \dots\dots\dots$
 $x = \dots\dots\dots$

Problems 6–7: Fill in the blank so the equation has:

6. One solution

$x(x - 2) = \dots\dots\dots$

7. No solutions

$(x + 3)(x - \dots\dots\dots) = -7$

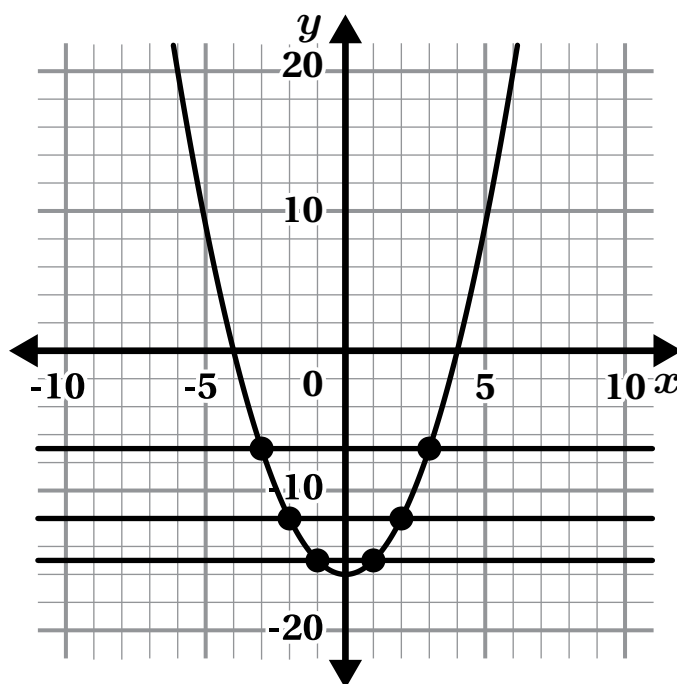
Problems 8–9: The graphs of $y = x^2 - 16$, $y = -15$, $y = -12$, and $y = -7$ all intersect at integer values.

8. Write the equations of three more lines that follow this property.

$y =$ _____

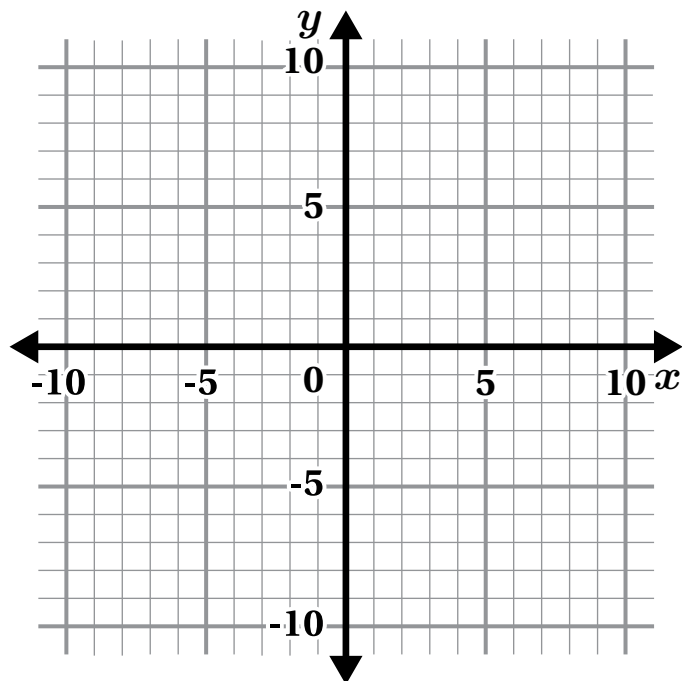
$y =$ _____

$y =$ _____



9. What pattern do you notice?

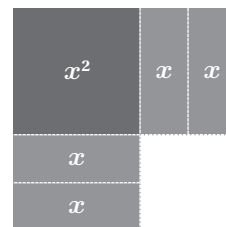
10. Solve $(x - 3)(x + 2) = 6$. Show or explain your thinking. Use the graph if it helps with your thinking.



Additional Practice

A1.8.11

1. The algebra tiles in the figure represent the expression $x^2 + 4x$. How many 1-tiles are needed to make the figure a square?



1-tile = 1

2. Which expression is a perfect square expression?

- A. $x(x - 4)$ B. $(x + 6)^2$
C. $x - 5^2$ D. $x^2 + 1$

3. Here is an incomplete quadratic expression: $x^2 - 14x + \dots$.

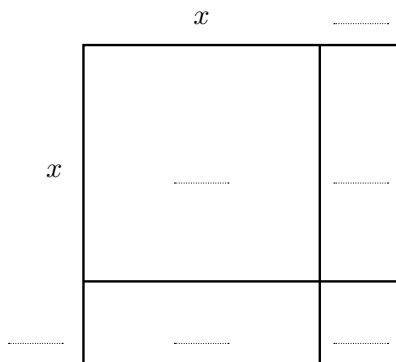
- a What value can be added to make the expression a perfect square expression?
- b Write the perfect square expression in factored form.

4. For each expression, determine the value that, when added to the expression, makes it a perfect square expression. Then write it in standard form and factored form.

Incomplete Expression	Perfect Square Expression in Standard Form	Perfect Square Expression in Factored Form
$x^2 + 12x + \dots$		
$x^2 - 18x + \dots$		
$x^2 + \dots + 121$		
$x^2 - 5x + \dots$		

5. Here is an incomplete quadratic expression: $x^2 + 9x + \dots$.

- a Complete the area diagram to turn the expression into a perfect square expression. What is the value of the missing term?
- b Write the perfect square expression in factored form.



6. Each of the following expressions written in standard form is an incomplete perfect square expression. Determine the missing value. Then match each expression in standard form with an equivalent expression in factored form.

Standard Form	Factored Form
a $x^2 + 14x + \dots$	$\dots (x - 3)^2$
b $x^2 + \dots x + 4$	$\dots (x + 7)^2$
c $x^2 - 6x + \dots$	$\dots (x + 2)^2$
d $x^2 - \dots x + 16$	$\dots (x - 4)^2$

7. Lin is changing the expression $x^2 + 16x$ so that it will be a perfect square expression. Her work is shown. Complete Lin's missing step.

Lin

$$x^2 + 16x = \boxed{}$$

$$= (x + 8)^2$$

8. Here is a quadratic expression: $x^2 - 30x$. Elena claims that 900 can be added to make it a perfect square expression. Shawn argues that the number 225 should be added instead. Who is correct? Explain your thinking.

Additional Practice

A1.8.12

1. What number can you add to make each expression a perfect square?

a $x^2 + 10x + 14$

b $x^2 - 8x - 3$

2. Add the number that would make the expression a perfect square. Then write an equivalent expression in factored form.

a $x^2 - 4x + \square$

b $x^2 + 10x + \square$

c $x^2 + 16x + \square$

.....

.....

.....

3. Jada wants to solve the quadratic equation $x^2 - 10x + 21 = 0$ by completing the square. Her work is shown. She says that there is one solution, $x = 5$. Is Jada correct? Explain your thinking.

Jada

$$x^2 - 10x + 21 = 0$$

$$x^2 - 10x + 21 + 4 = 0$$

$$x^2 - 10x + 25 = 0$$

$$(x - 5)^2 = 0$$

4. Solve each equation by completing the square.

a $x^2 - 4x - 3 = 2$

b $x^2 + 8x + 3 = -9$

c $x^2 - 14x + 54 = 5$

d $7 = x^2 + 6x - 20$

5. Which of the following could be the first step in solving the quadratic equation $x^2 + 8x - 20 = 0$ by completing the square?

A. $x^2 + 8x - 16 = 20$ B. $x^2 + 8x - 20 + 36 = 0 + 36$
C. $x^2 + 8x - 20 = -20$ D. $x^2 + 8x - 20 + 16 = 0$

6. Priya wants to solve the quadratic equation $x^2 + 12x + 11 = 0$ by completing the square. Her work is shown. She says that the solutions are $x = 19$ and $x = -31$. Which of the following statements is true?

A. Priya's work and solutions are correct.
B. Priya made an error in the second line by adding 25 to each side of the equation.
C. Priya made an error in the third line by incorrectly writing the expression in factored form.
D. Priya made an error in the fourth line by not taking the square root of 25.

Priya

$$x^2 + 12x + 11 = 0$$

$$x^2 + 12x + 36 = 25$$

$$(x + 6)^2 = 25$$

$$x + 6 = 25 \text{ and } x + 6 = -25$$

$$x = 19 \text{ and } x = -31$$

7. Here is the quadratic equation $x^2 + 16x + 38 = 23$. Bard solves the equation by completing the square and says there is only one solution, $x = -1$. Is Bard correct? Explain your thinking.

Additional Practice

A1.8.14

1. Which of the following is the solution to any quadratic equation $ax^2 + bx + c = 0$?

- A. $x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$ B. $x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$ C. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$
 D. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ E. $y = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

Problems 2–5: Identify the values for a , b , and c in each of the quadratic equations below.

	Equation	a -term	b -term	c -term
2.	$-2x^2 - 7x + 5 = 0$			
3.	$8x^2 = 2x + 4$			
4.	$4x^2 = 12$			
5.	$2x(x - 5) = 0$			

6. Use the quadratic formula to show the equation $x^2 + 6x + 10 = 0$ has no solutions. Show or explain how you know.

$$x^2 + 6x + 10 = 0$$

$$a = 1 \quad b = 6 \quad c = 10$$

Problems 7–10: Here is the equation $x^2 - 8x = -7$.

7. Write the equation in standard form.

8. Identify the values for a , b , and c .

$a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

9. Substitute the values for a , b , and c into the quadratic formula. (You do not need to perform any operations.)

10. Show or explain how the expression you wrote is related to solving $x^2 - 8x = -7$ by completing the square.

Additional Practice

A1.8.15

1. Each equation is in the form $ax^2 + bx + c = 0$. Determine the values of a , b , and c .

a $2x^2 + 5x + 1 = 0$

b $-4x^2 + 3x + 6 = 0$

c $x^2 - 7x + 4 = 0$

d $-x^2 + 10x - 8 = 0$

e $24 + 2x - x^2 = 0$

f $\frac{3}{4}x^2 - 8x = \frac{1}{4}$

g $x^2 = 100$

h $5x - 2x^2 = -19$

2. What are the values of x if $x = \frac{7 \pm \sqrt{49 - (-72)}}{4}$?

A. 0 and $-\frac{9}{2}$

B. 0 and 1

C. -1 and $\frac{9}{2}$

D. $-\frac{9}{2}$ and 1

3. Determine whether the solutions for each quadratic equation are correct.

a Equation: $x^2 - x - 56 = 0$

Solutions: $x = -7$ and $x = 8$

b Equation: $x^2 + 4x - 45 = 0$

Solutions: $x = -5$ and $x = 9$

c Equation: $2x^2 - 8x + 3 = 0$

Solutions: $x = \frac{4 + \sqrt{10}}{2}$ and $x = \frac{4 - \sqrt{10}}{2}$

4. Solve each quadratic equation using the quadratic formula. Show your thinking.

a $x^2 + 5x - 36 = 0$

b $2x^2 + 9x - 35 = 0$

c $2x^2 - 16x + 30 = 0$

d $x^2 + 8x + 4 = 0$

5. Solve each quadratic equation using any strategy.

a $x^2 - 2x - 63 = 0$

b $2x^2 + 18x + 40 = 0$

c $4x^2 - 45x + 11 = 0$

d $3x^2 + 15x - 2 = 0$

6. Diego is solving $5x^2 - x - 8 = 0$ using the quadratic formula. His work is shown. Are his solutions correct? Explain your thinking.

Diego

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(5)(-8)}}{2(5)}$$
$$x = \frac{1 \pm \sqrt{-1 + 160}}{10}$$
$$x = \frac{1 + \sqrt{159}}{10} \text{ and } x = \frac{1 - \sqrt{159}}{10}$$

Additional Practice

A1.8.16

1. The function $g(t) = -4.9t^2 + 40t + 6$ models the height, in meters, of an apple t seconds after it has been launched into the air from a toy cannon.
 - a Write an equation to determine when the apple will hit the ground.
 - b Determine when the apple hits the ground. Round to the nearest thousandth.

2. Two objects are launched into the air. The function $h(t) = 45 + 130t - 16t^2$ models the height, in feet, of Object A t seconds after it has been launched. The function $f(t) = 18 + 70t - 16t^2$ models the height, in feet, of Object B t seconds after it has been launched.
 - a After 5 seconds have passed, which object is still in the air?
 - b Write an equation for each object to determine when it will hit the ground.
 - c Determine when each object hits the ground. Round to the nearest thousandth.

3. A picture is 10 inches wide, 15 inches long, and has a frame x inches thick surrounding it. The equation $(10 + 2x)(15 + 2x) = 336$ represents the total area, in square inches, of the picture in its frame.
 - a Rewrite the equation $(10 + 2x)(15 + 2x) = 336$ in standard form.
 - b Determine the thickness of the frame.

- 4.** A picture is 8 inches wide and 10 inches long. It has a frame x inches thick surrounding it. The equation $(8 + 2x)(10 + 2x) = 195$ represents the total area, in square inches, of the picture and the frame.
- a** Rewrite the equation $(8 + 2x)(10 + 2x) = 195$ in standard form.
 - b** Solve your equation. What do the solutions represent?
- 5.** A ball is thrown up into the air. Its height in inches is modeled by the function $h(t) = -16t^2 + 28t + 3$, where t is the time after the ball is thrown, measured in seconds. Solve the equation $-16t^2 + 28t + 3 = 0$. What do the solutions tell you about the ball?
- 6.** A projectile is launched into the air and its height above the ground, in feet, is modeled by the function $f(t) = 6 + 32t - 16t^2$, where t is the number of seconds since the projectile was launched.
- a** What are the solutions to the equation $0 = 6 + 32t - 16t^2$?
 - b** Do both of the solutions have meaning in this context? Explain your thinking.

Additional Practice

A1.8.17

1. Select *all* the expressions that are rational.

☐ A. $\frac{1}{4} \cdot \sqrt{9}$

☐ B. $\sqrt{5} + \sqrt{9}$

☐ C. $2.6 + 5.3$

☐ D. $-\sqrt{7} + \sqrt{7}$

☐ E. $\sqrt{16} \cdot \sqrt{4}$

☐ F. $6 + \sqrt{6}$

2. Select *all* the following numbers that will result in a rational number when multiplied by $\sqrt{2}$.

☐ A. 2

☐ B. 8

☐ C. $\frac{1}{2}$

☐ D. $\sqrt{2}$

☐ E. $\sqrt{4}$

☐ F. $\sqrt{8}$

3. Determine whether each of the following statements are *always*, *sometimes*, or *never true*. Explain your thinking.

a Two rational numbers added together will be rational.

b Multiplying an irrational number by another irrational number will result in an irrational number.

c The sum of a rational number that and an irrational number is rational.

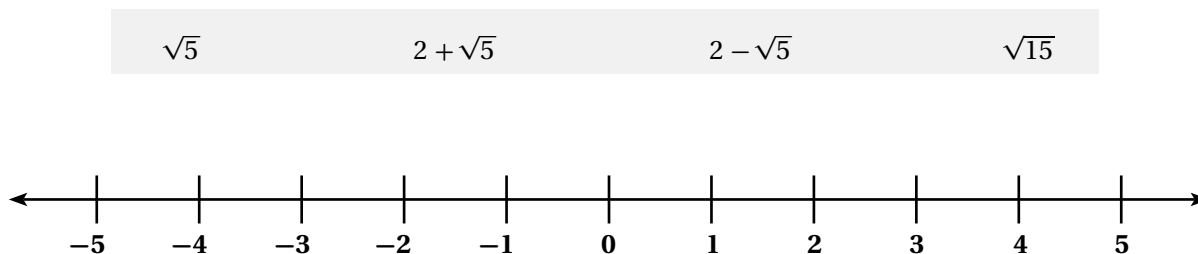
4. Consider the statement: *The product of a rational number and an irrational number is irrational.* Which example shows this statement is false?

- A. $\sqrt{7} \cdot \sqrt{7} = 7$
 B. $2 \cdot \sqrt{2} = 2\sqrt{2}$
 C. $\sqrt{4} \cdot \sqrt{16} = 8$
 D. $\sqrt{7} \cdot 0 = 0$

5. Here are the solutions to some quadratic equations. Decide if the solutions are rational or irrational.

a	$5 \pm \sqrt{25}$	Rational	Irrational
b	$-8 \pm \sqrt{6}$	Rational	Irrational
c	$\frac{3}{4} \pm \sqrt{\frac{70}{7}}$	Rational	Irrational
d	2 ± 9	Rational	Irrational
e	$3 \pm \sqrt{2}$	Rational	Irrational
f	8 ± 0.5	Rational	Irrational

6. Here is a number line with four points on it. Estimate each expression and plot it on the number line.



7. Consider the expressions $\sqrt{2} + a$ and $b\sqrt{2}$. Determine a value of a and b so that both expressions will result in a rational number.